

From Audio-Visual Objects to Audio-Visual Spaces

Ruohan Gao

University of Maryland, College Park



Visual Generation is Advancing Rapidly



NeRF, Mildenhall et al, ECCV 2020



3DGS, Kerbl et al, SIGGRAPH 2023

What we can do visually (today)?

- Photorealistic 3D objects and scenes
- Novel-view synthesis
- Real-time view updates
- Semantics & lighting cues

.....

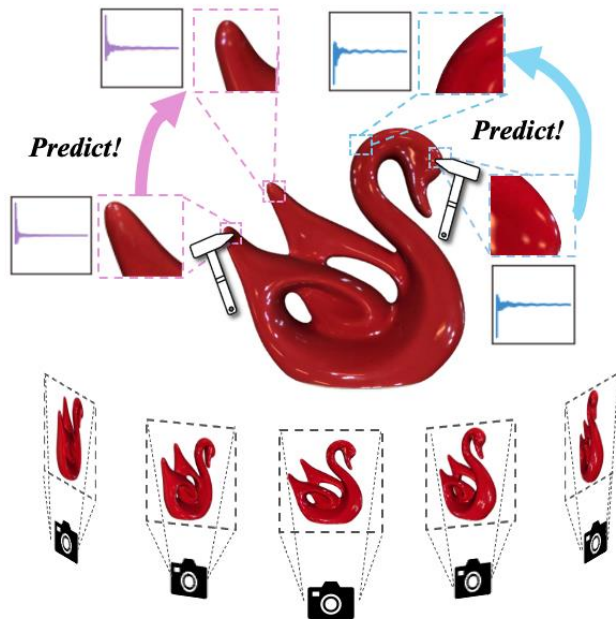
An aerial photograph of a city, likely Paris, featuring a large cathedral with a prominent dome and a modern building with a glass facade. The image is overlaid with a semi-transparent white rectangle containing the text "The Sound is Missing!".

The Sound is Missing!



Today's Talk

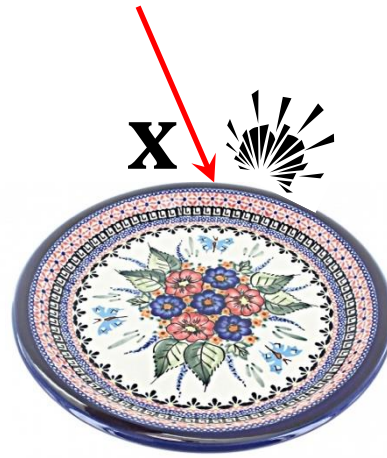
Audio-Visual Objects



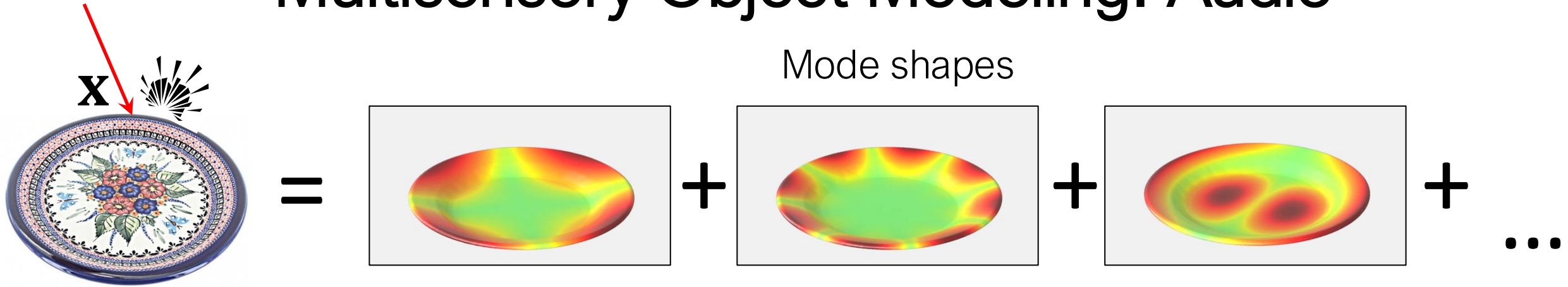
Audio-Visual Spaces



Impact Sound Modeling



Multisensory Object Modeling: Audio



Linear model analysis (established theory)

$$M\ddot{\mathbf{u}} + D\dot{\mathbf{u}} + K\mathbf{u} = \mathbf{f}(t)$$



Mass



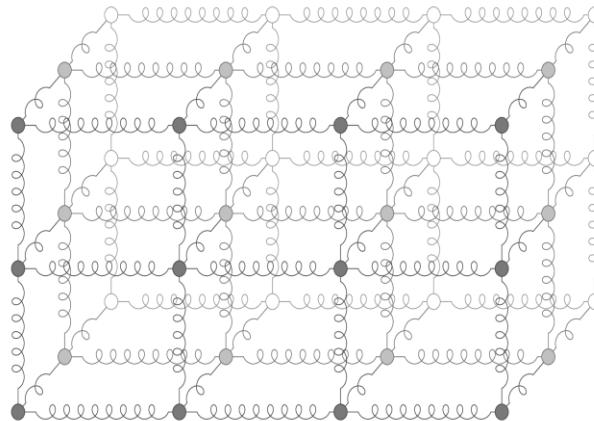
Damping



Stiffness

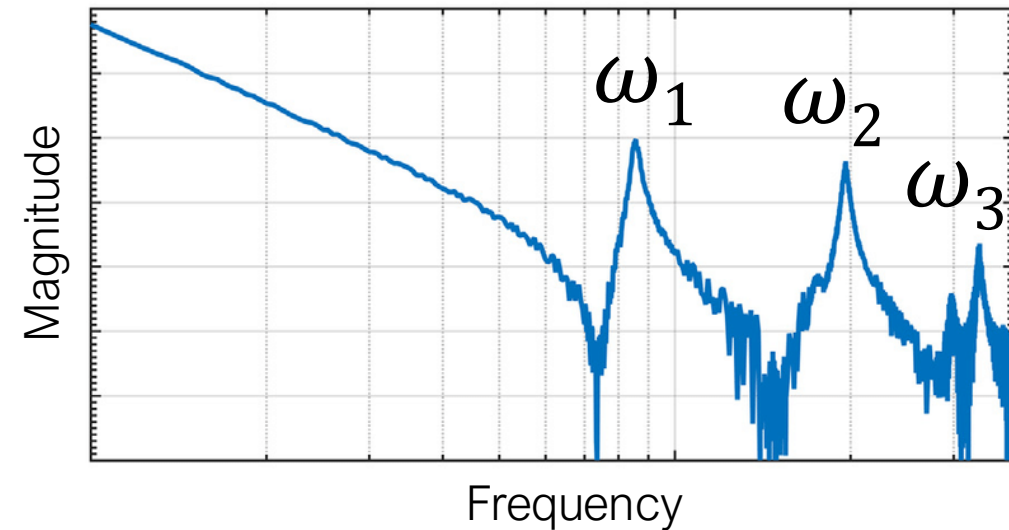
Rayleigh damping:

$$D = \alpha M + \beta K$$

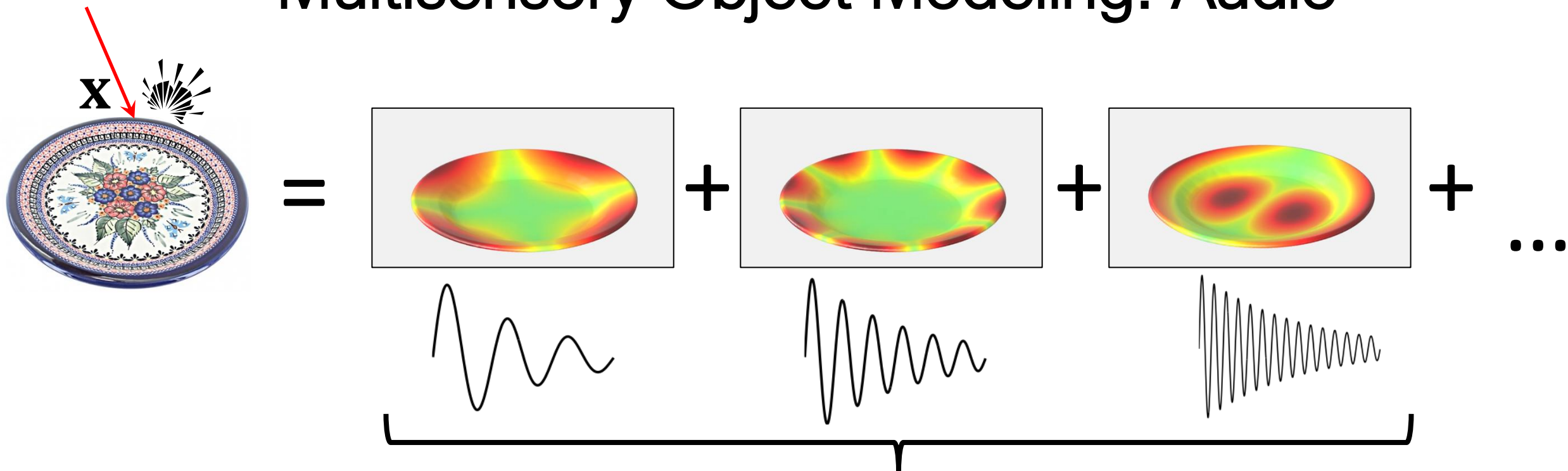


Many DOF mass-spring system

Frequency response



Multisensory Object Modeling: Audio

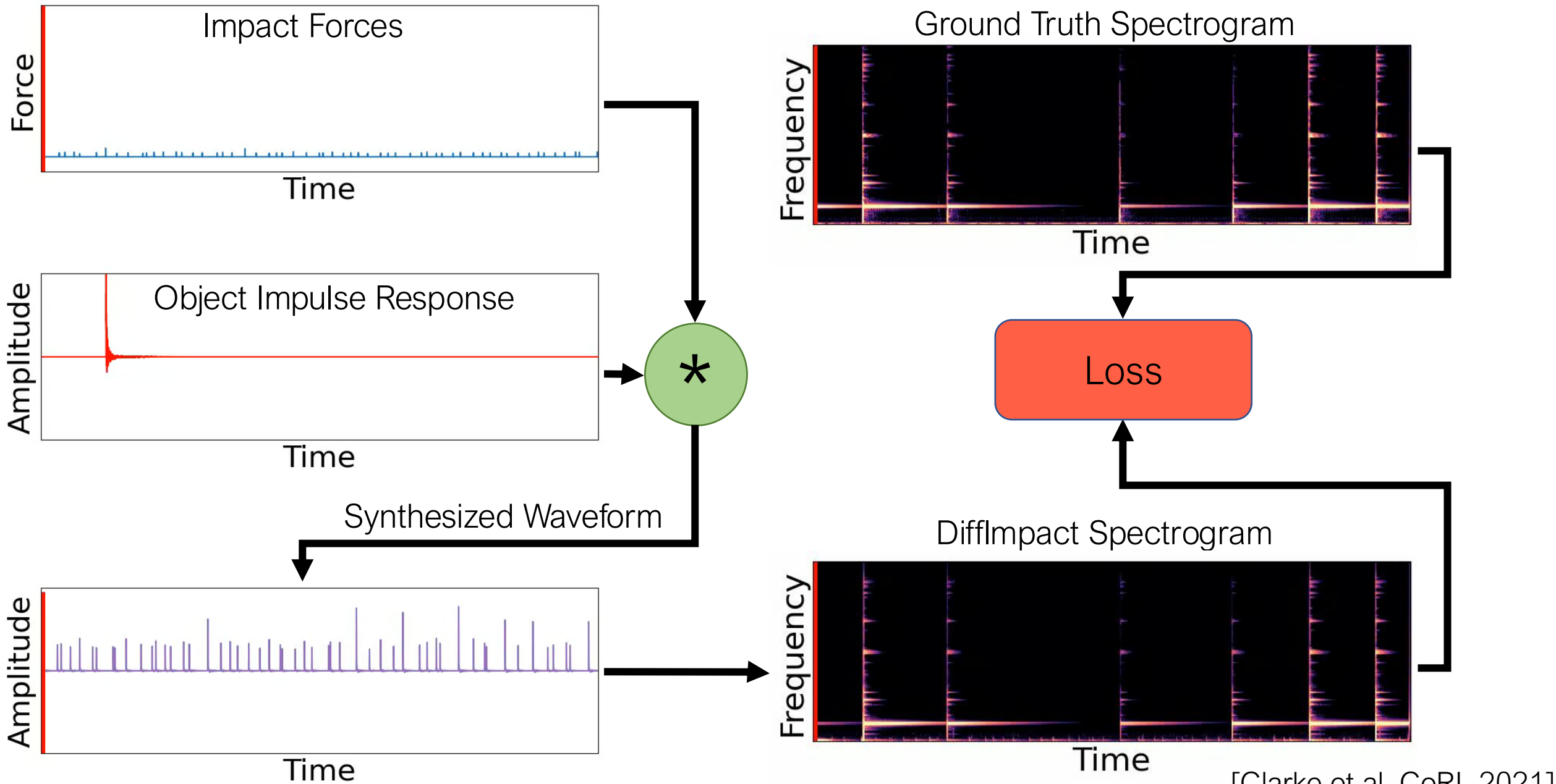


Object impulse response:

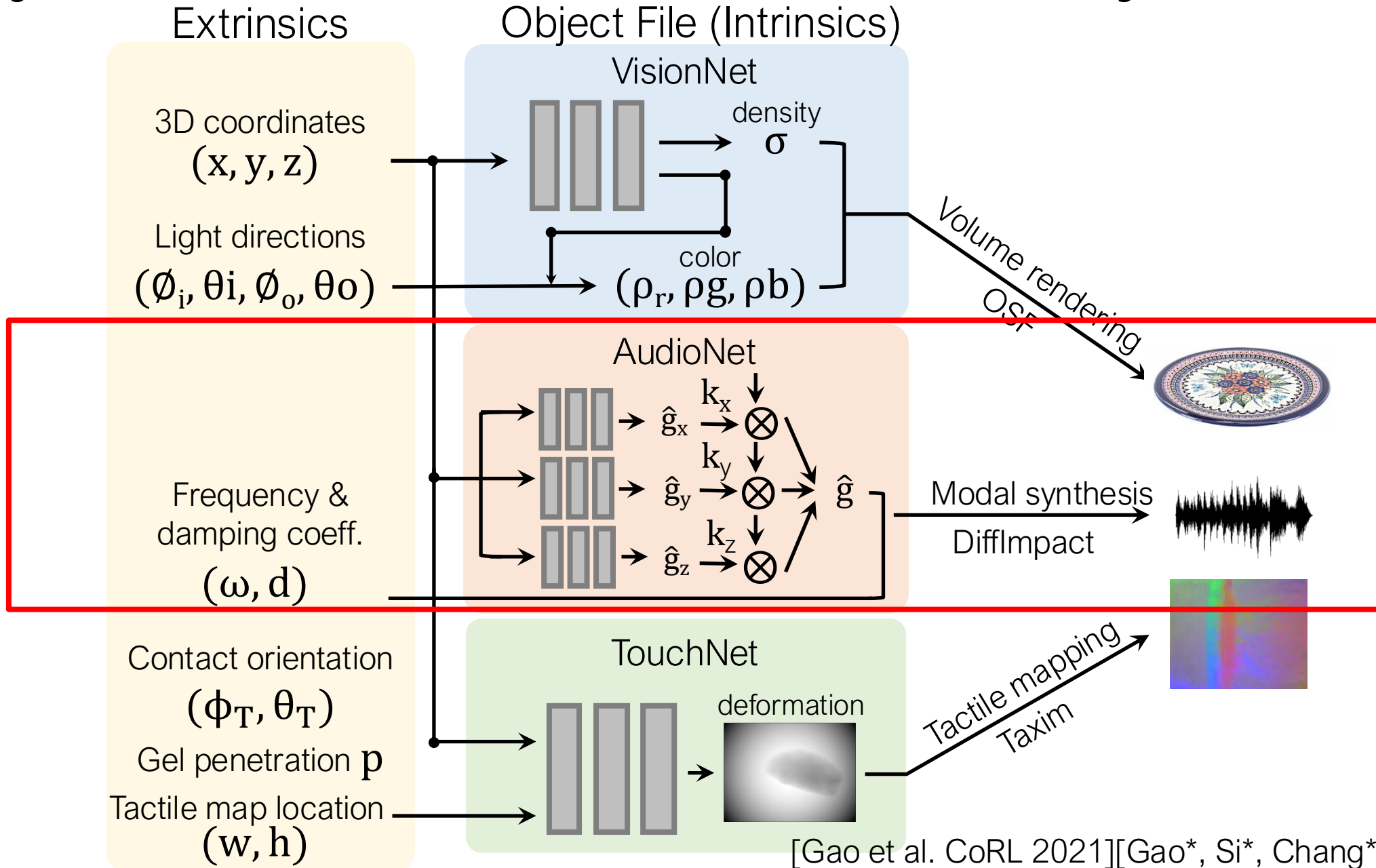
$$sound(t) = \sum_{i=1}^N g_i e^{-d_i t} \sin(2\pi\omega_i t)$$

↑ ↑ ↑
Gain Damping Frequency

Differentiable Impact Sound Modeling



ObjectFolder: A Dataset of Multisensory Neural Objects



ObjectFolder Audio Example: YCB Mug



Material type: Steel

Scale: 0.082 m



ObjectFolder Audio Example: Porcelain Pitcher



Material type: Ceramic

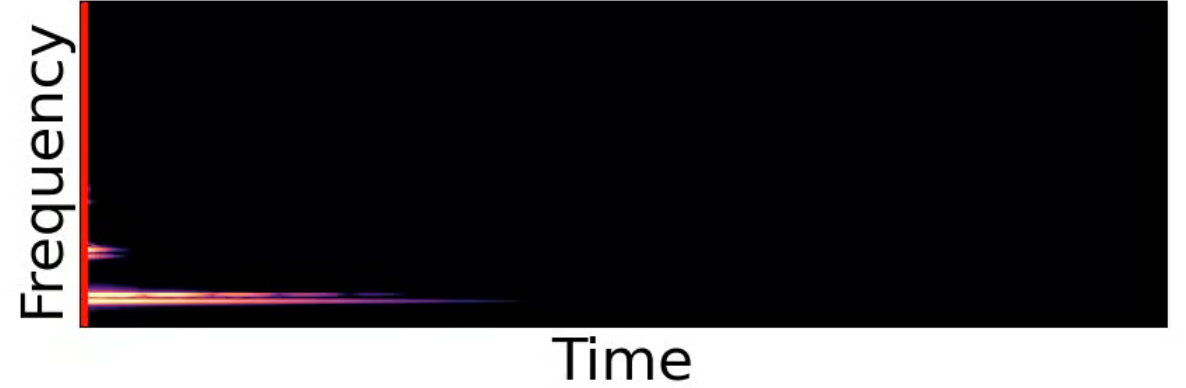
Scale: 0.278 m



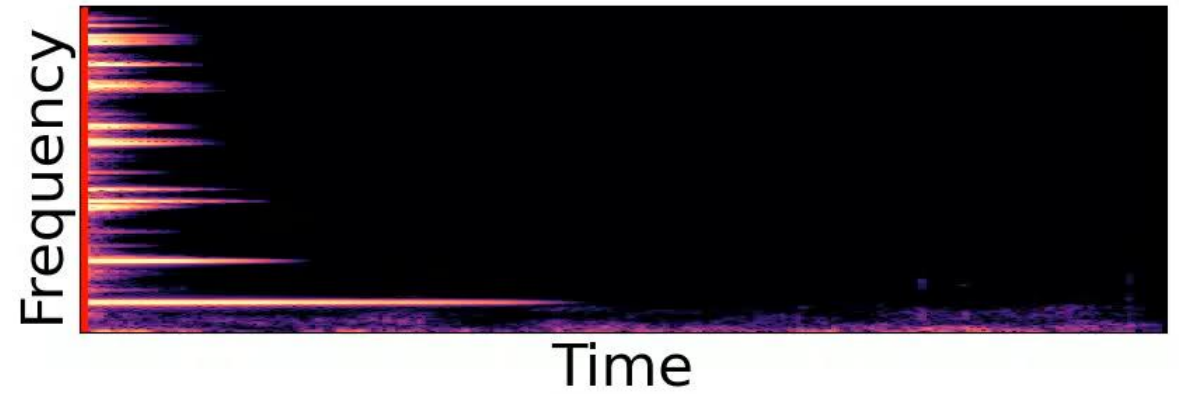
Simulation vs. Real



Simulation:



Real recording:

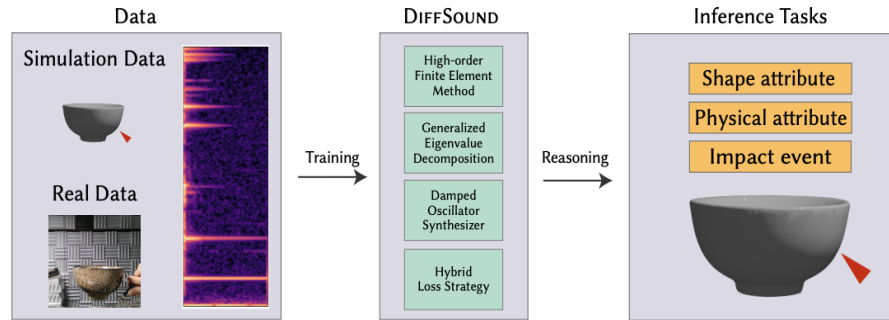


Acoustic Data in ObjectFolder Real

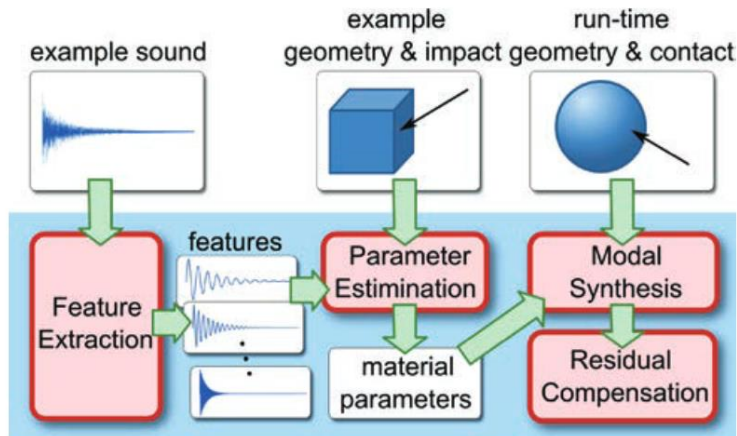


Prior Work on Object Impact Sound Modeling

Physics-based methods



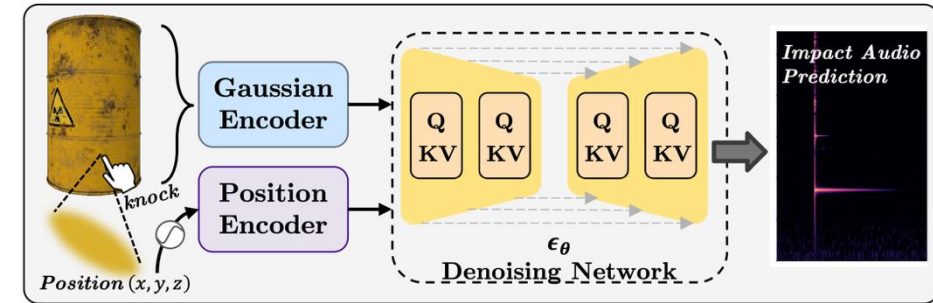
[Jin et al., SIGGRAPH 2024]



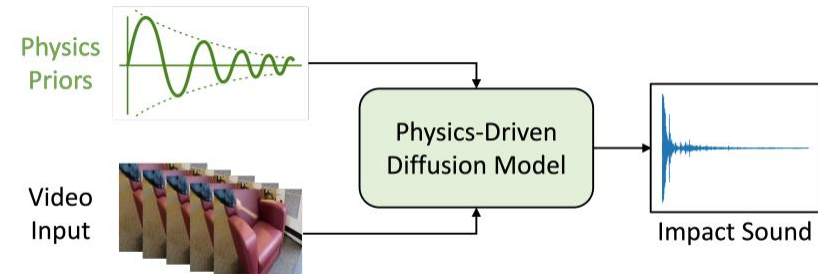
[Ren et al., TOG 2013]

- Computationally expensive
- Sensitive to errors in geometry and material parameters

Data-driven methods

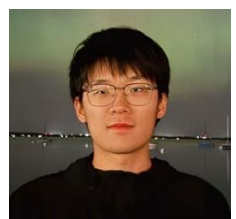


[Wang et al., ACMMM 2025]



[Su et al., CVPR 2023]

- Requires large-scale training data
- Limited physical interpretability



Zisen Shao



Zihao Wei

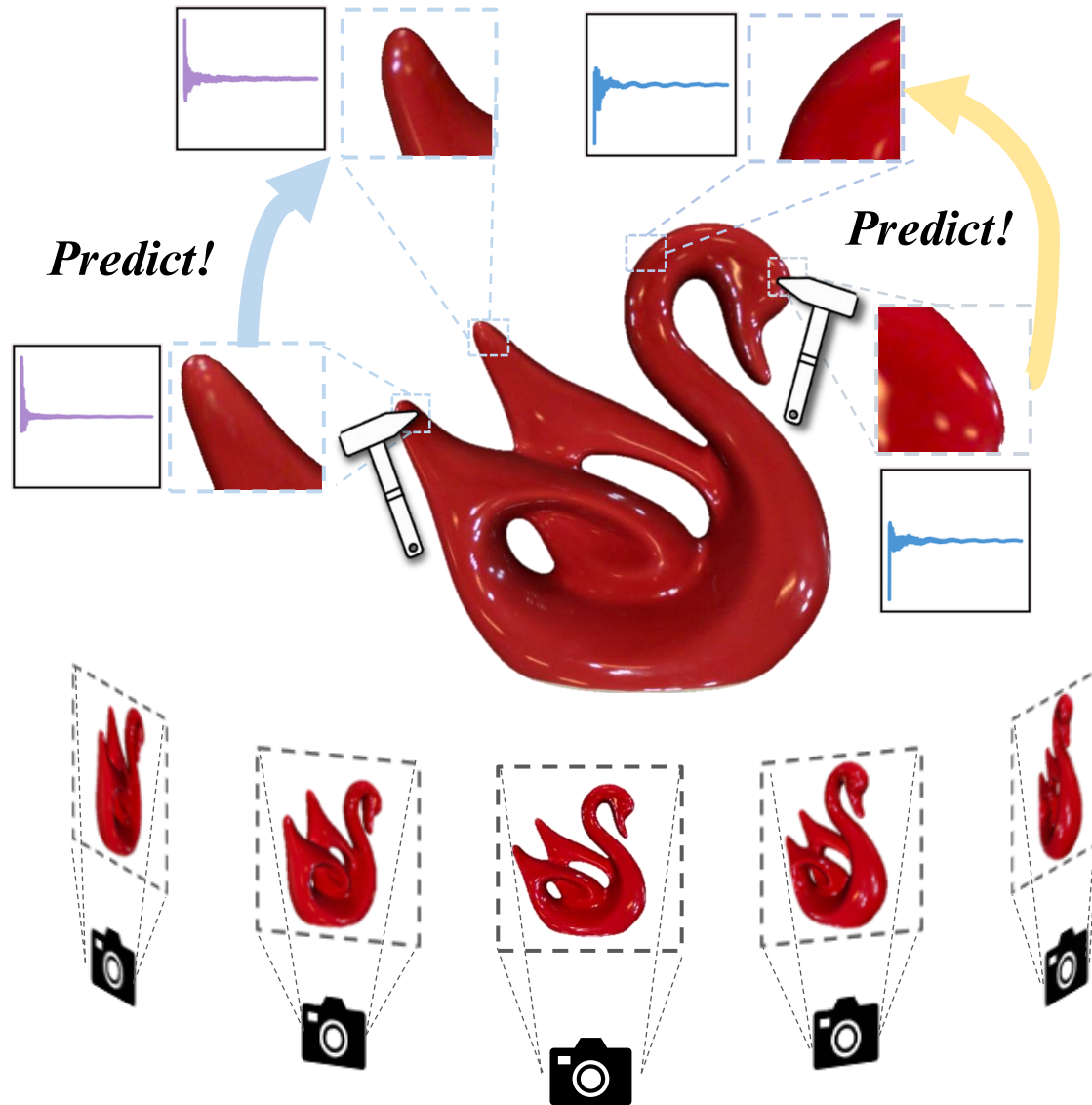
Our Key Observations

Visual and geometrical priors help impact sound prediction

Visually similar object regions exhibit similar vibration patterns!



Visual Cues Inform Impact Sound Prediction



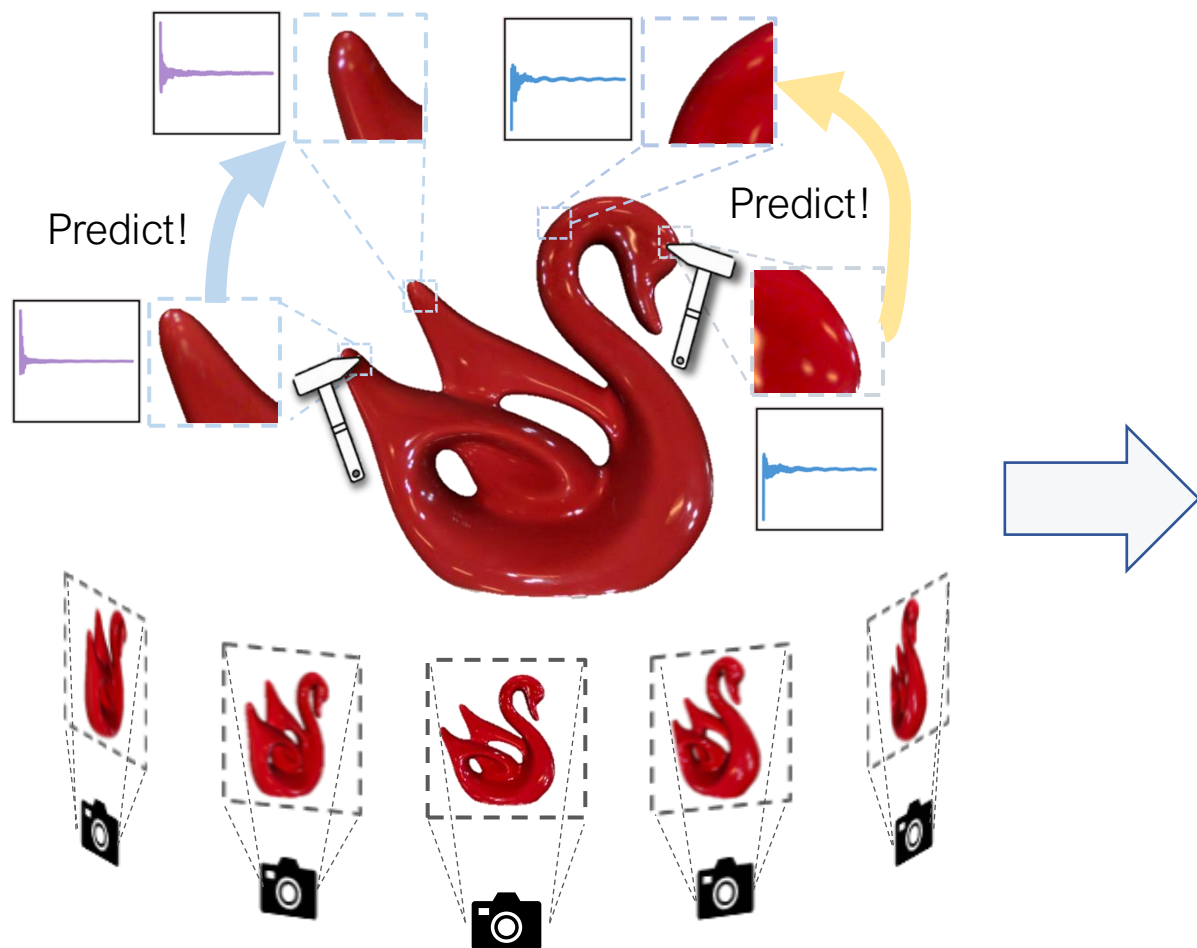
Our Key Observations

Visual and geometrical priors help impact sound prediction

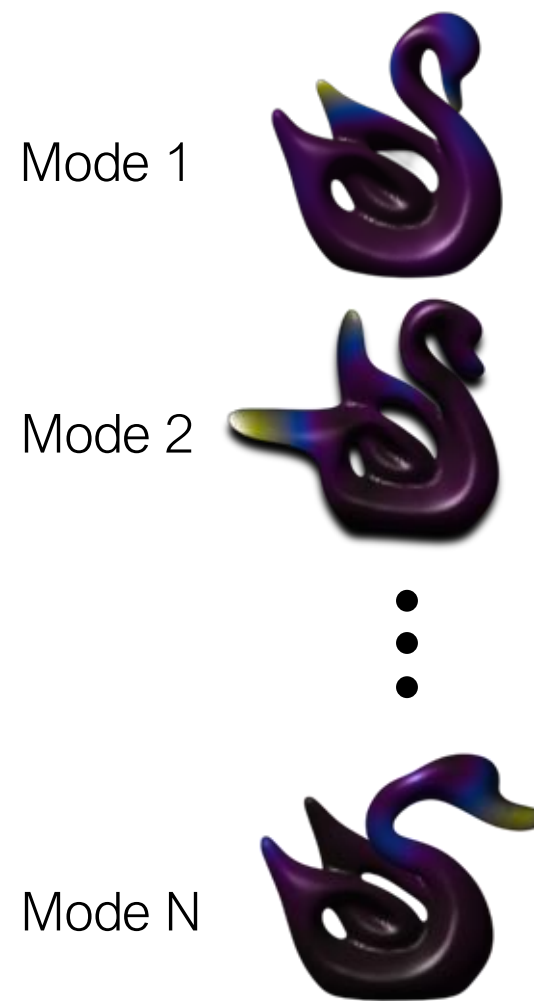
Physics-based modal sound representation that is data-efficient

Physics-based Modal Sound Representation

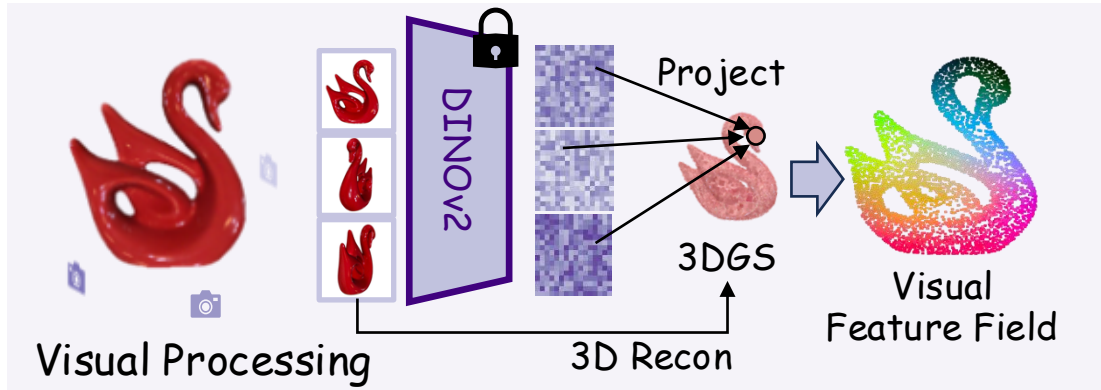
Audio-Visual Priors



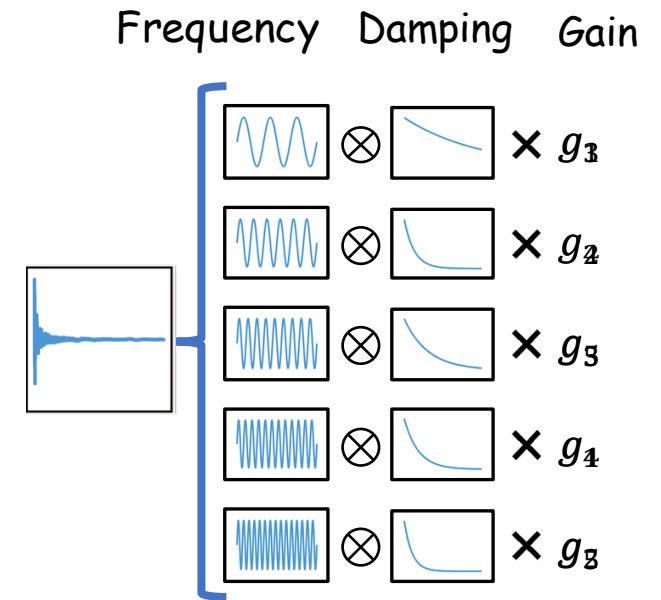
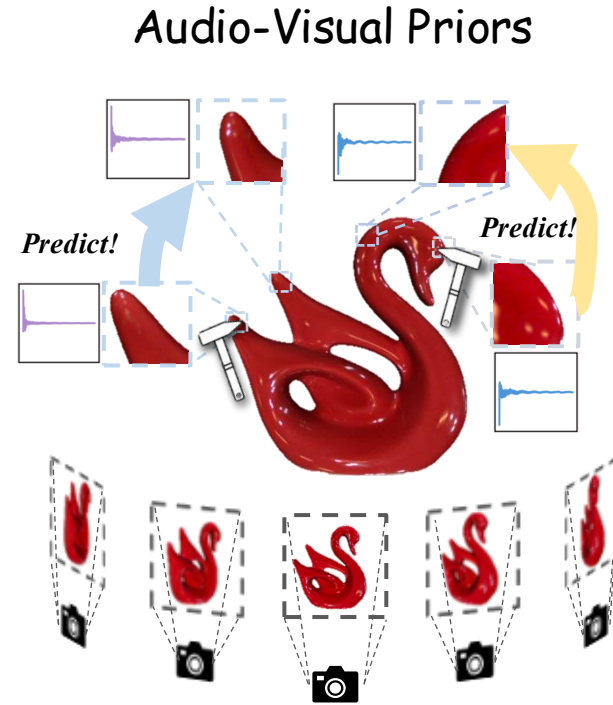
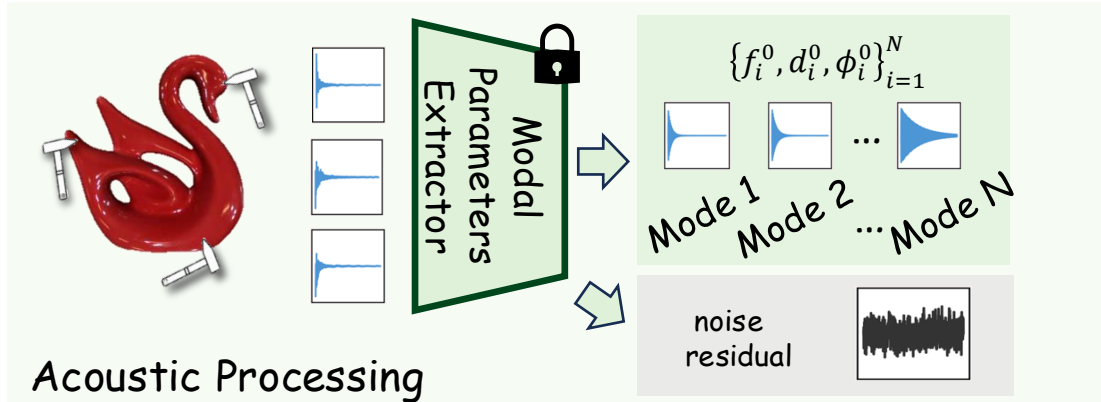
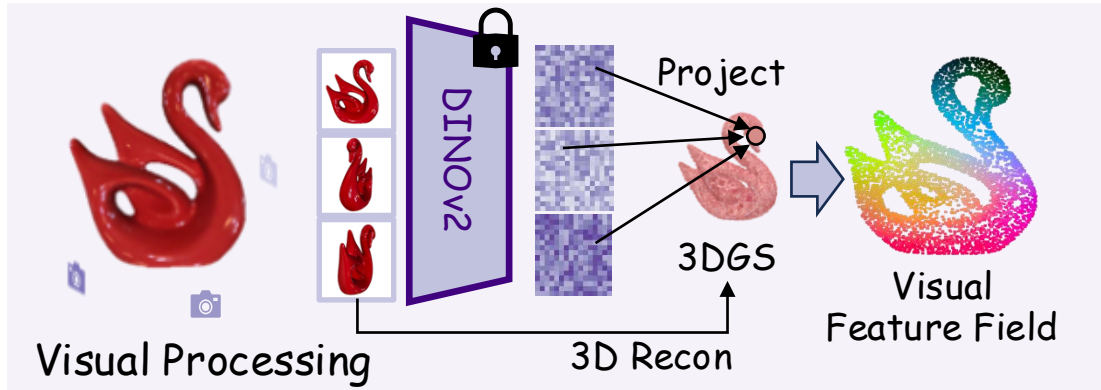
Vibration Modes



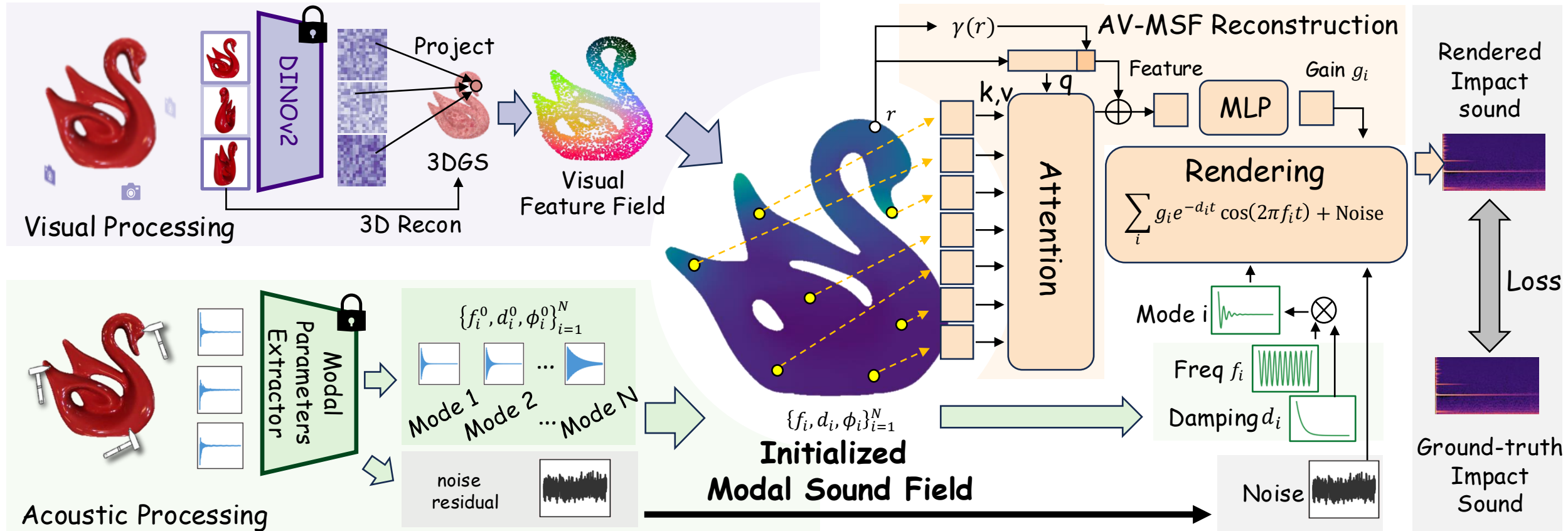
Audio-Visual Modal Sound Field Framework



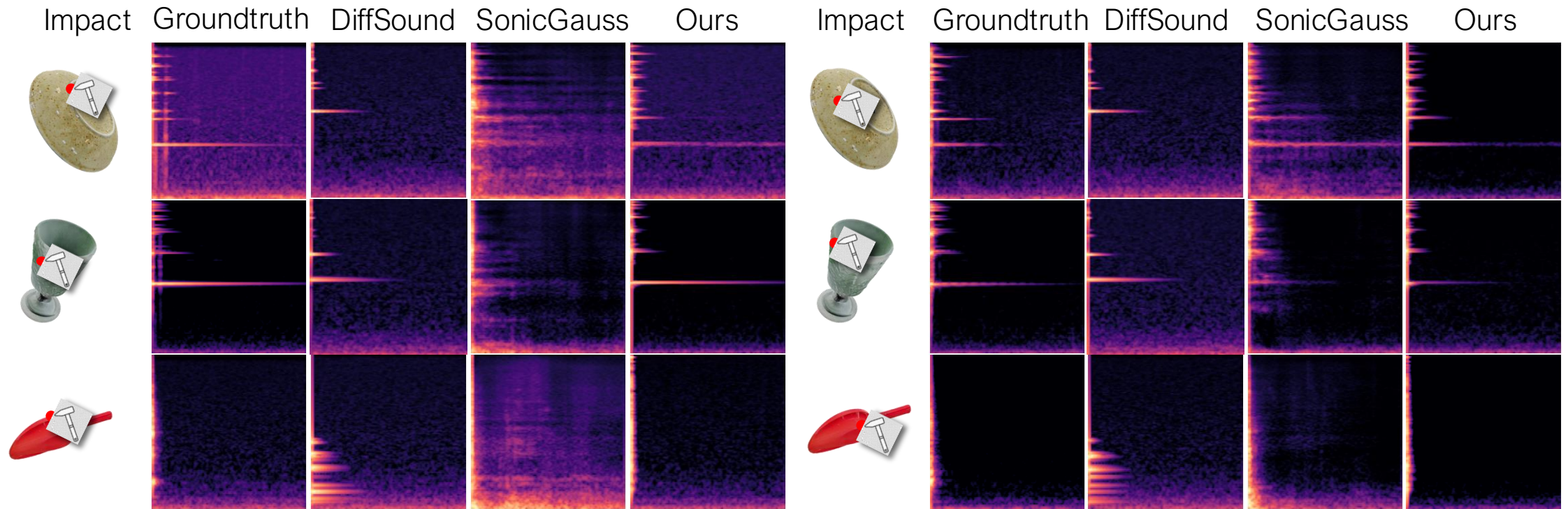
Audio-Visual Modal Sound Field Framework



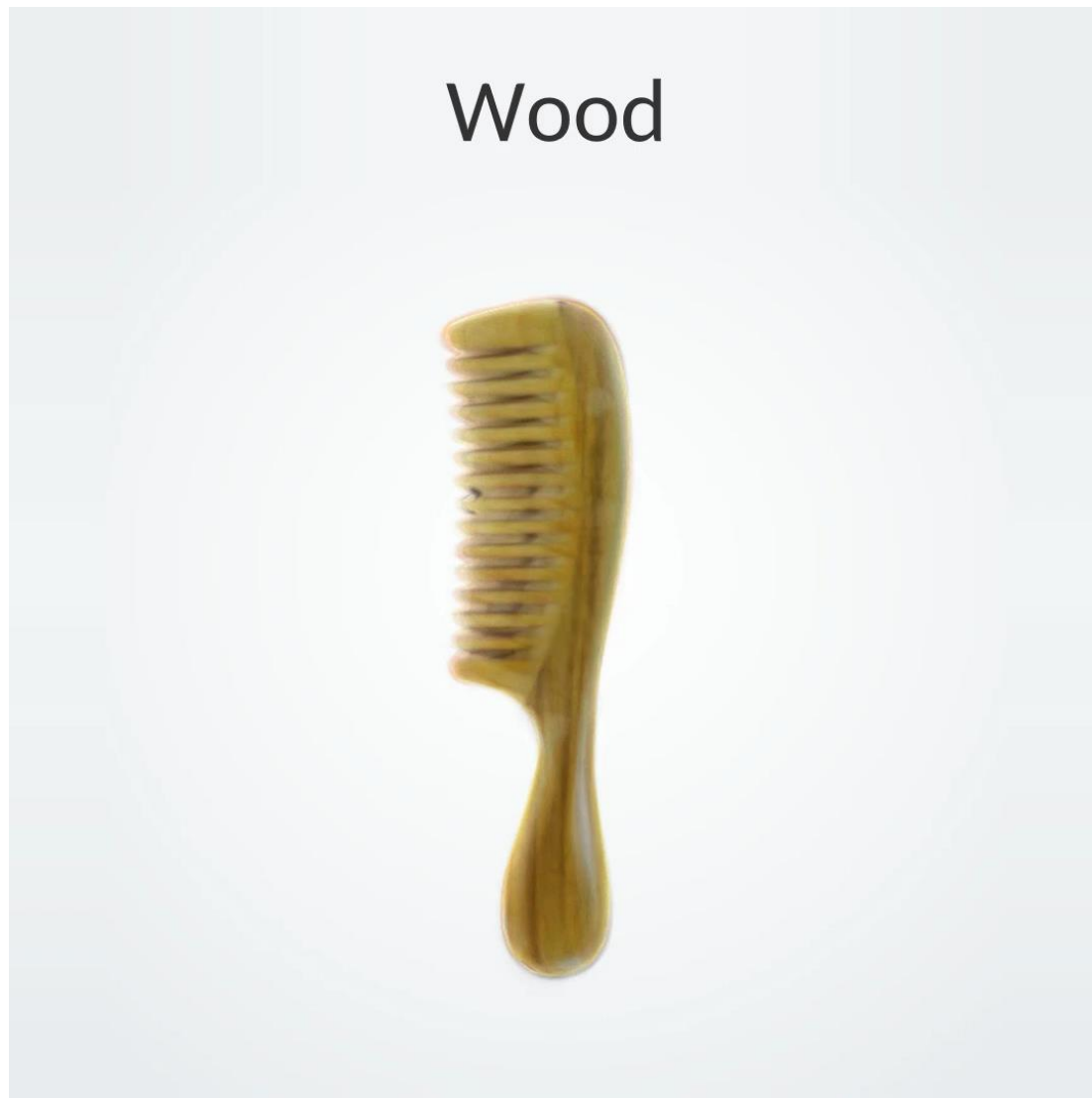
Audio-Visual Modal Sound Field Framework



Qualitative Results on Impact Sound Rendering



Impact Sound Rendering at Novel Positions



trained on ~6 shots

[Shao*, Wei*, Jin, Gao, ECCV 2026]

Quantitative Results on Impact Sound Rendering

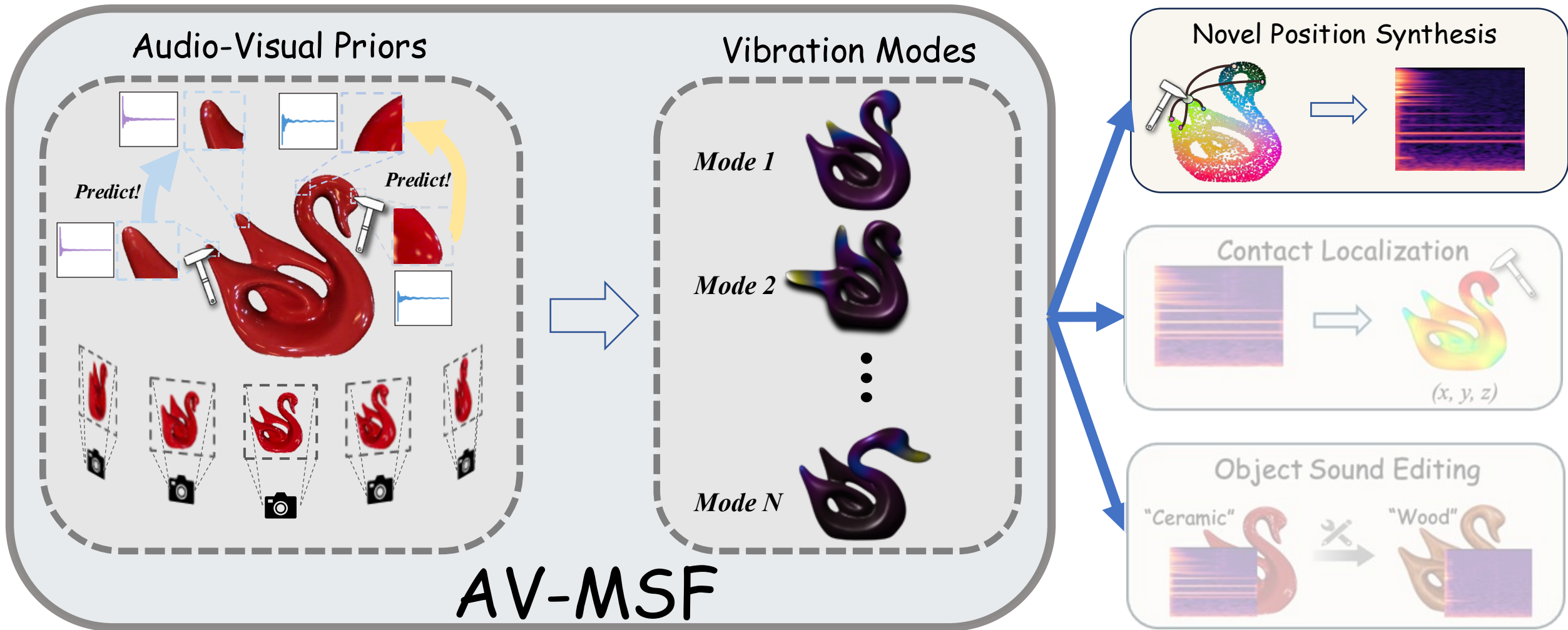
ObjectFolder Real: Gao et al. CVPR'23
ReallImpact: Clarke et al. CVPR'23



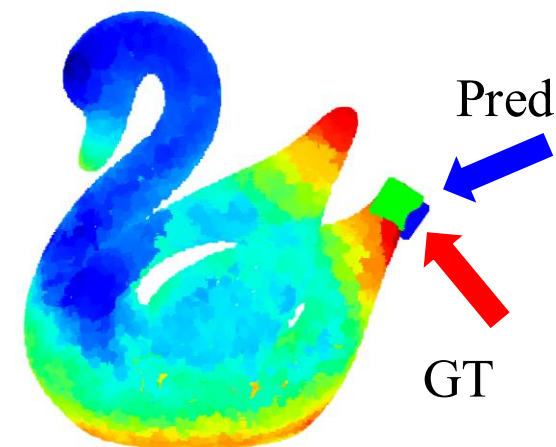
Method	OBJECTFOLDER REAL				REALIMPACT			
	L1	L1 Log	ENV	CDPAM	L1	L1 Log	ENV	CDPAM
WHITE NOISE	3.774	6.859	0.305	1.35e-3	3.789	7.258	0.308	1.23e-3
RANDOM IMPACT	0.035	1.367	0.019	2.47e-4	0.039	1.582	0.022	2.71e-4
KNN	0.014	0.930	0.014	1.53e-4	0.026	1.036	0.017	2.25e-4
DIFFSOUND [19]	0.031	1.298	0.030	2.53e-4	0.029	1.533	0.024	2.39e-4
SONICGAUSS [42]	0.033	1.281	0.018	2.01e-4	0.039	1.580	0.025	2.43e-4
Ours	0.013	0.951	0.014	1.35e-4	0.021	0.996	0.017	2.16e-4

Best impact sound prediction results compared to
both physics-based and data driven methods.

Applications of Object Impact Sound Synthesis



Contact Localization with Impact Sound



Physically Grounded Object Sound Editing



Original: Ceramic

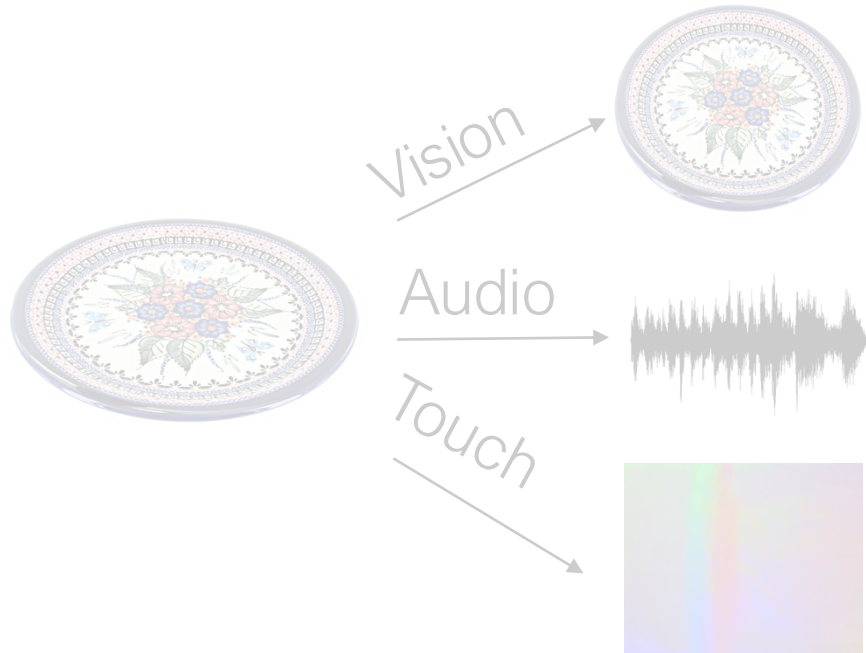
A deep, solid, dull
wooden thud



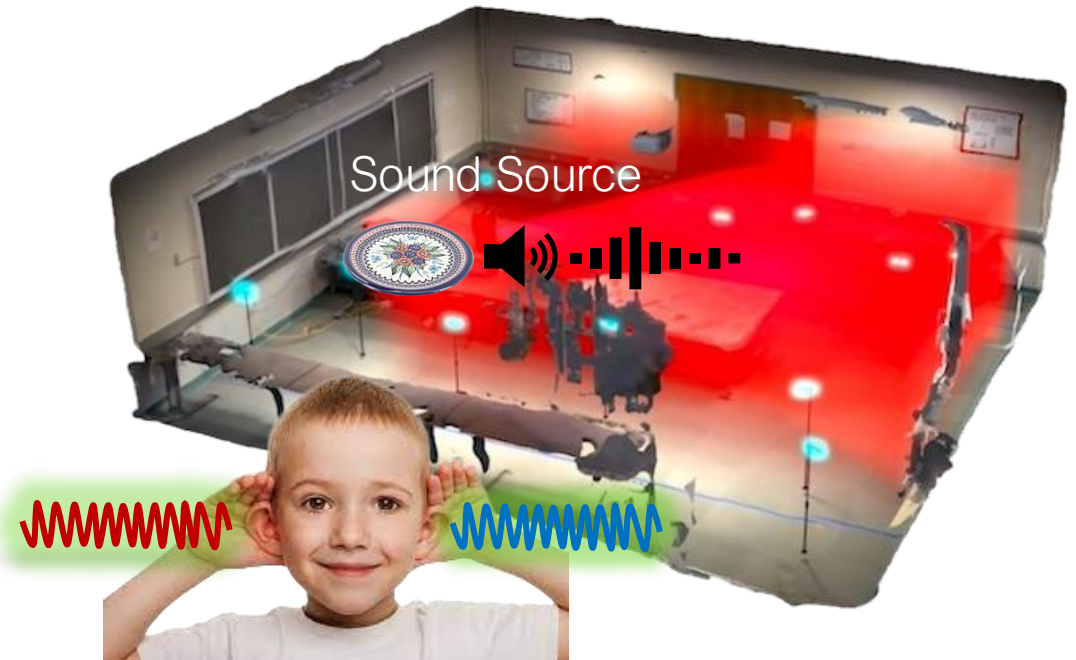
Target: Wood

Today's Talk

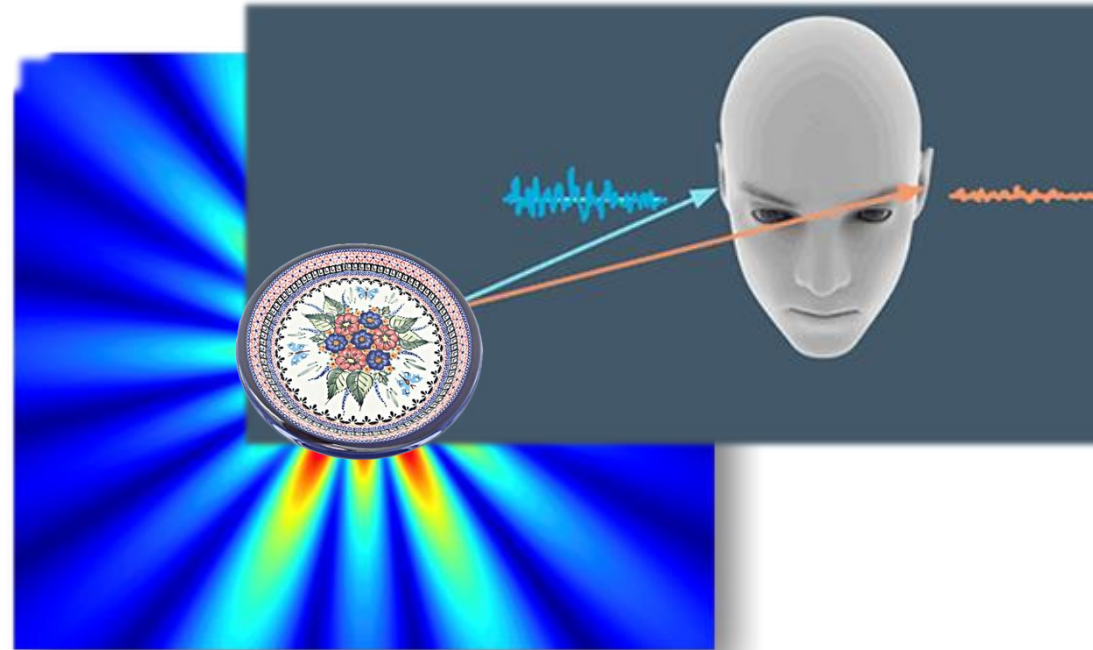
Audio-Visual Objects

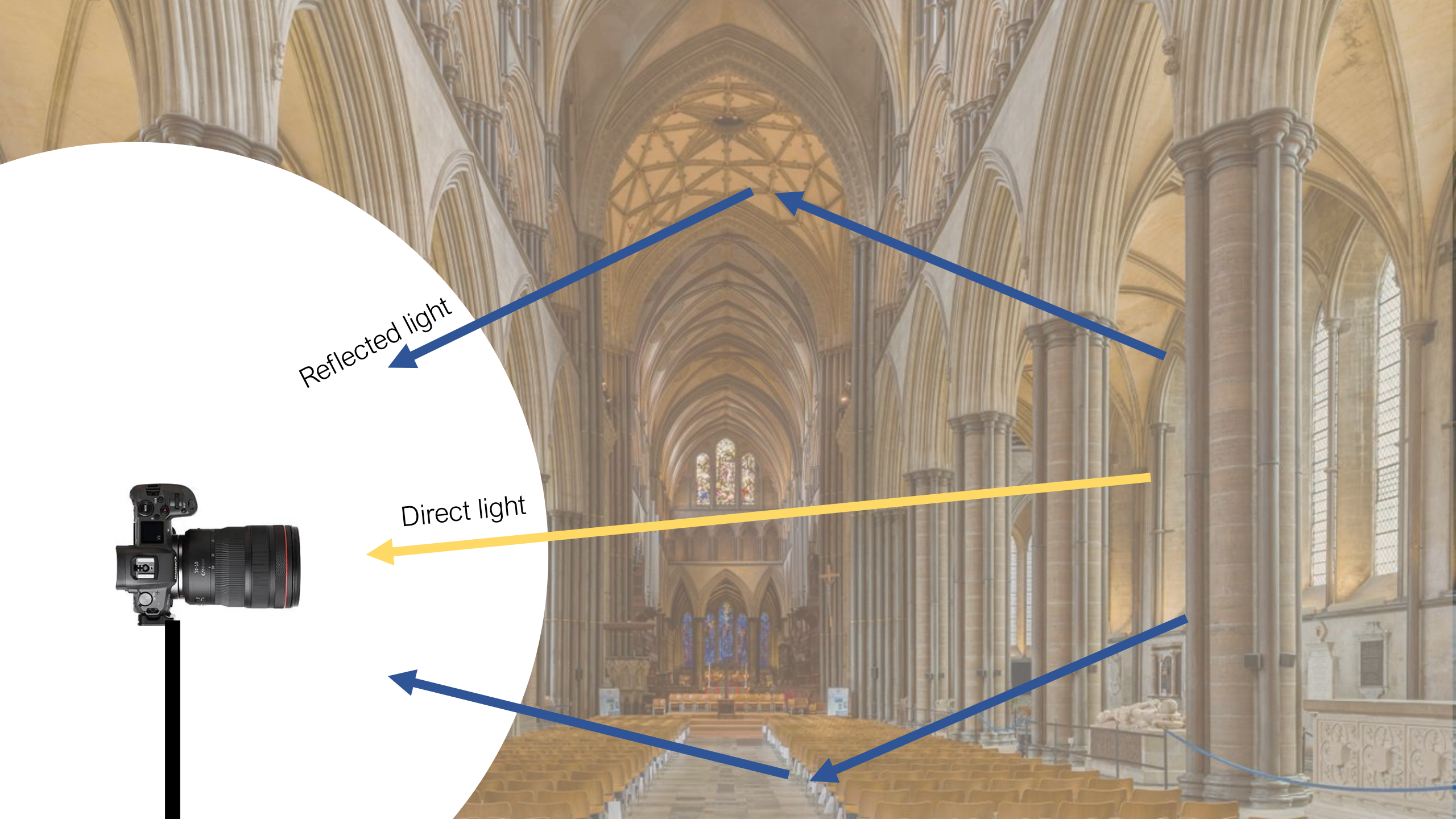


Audio-Visual Spaces



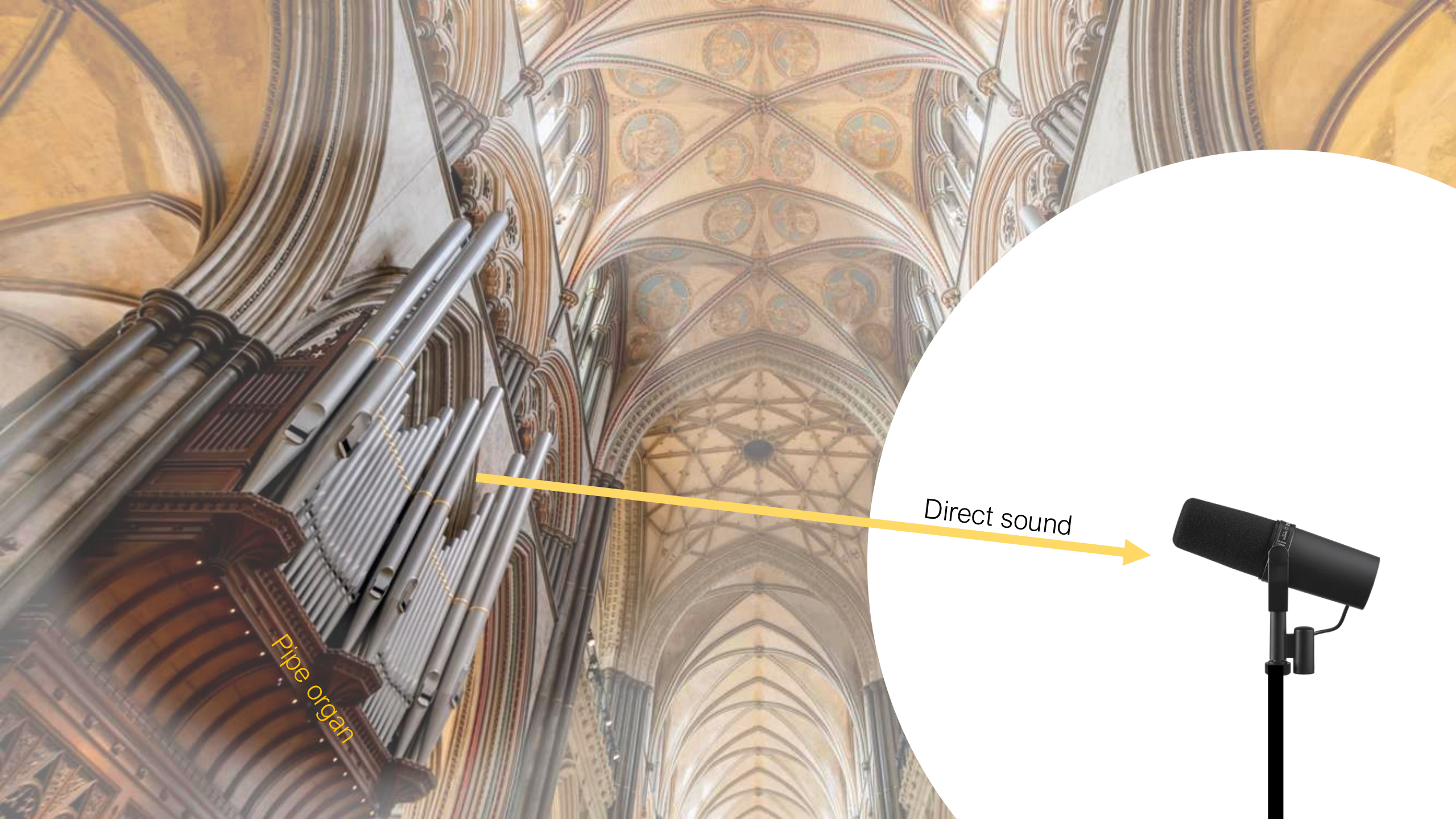
Human Perception of Spatial Audio





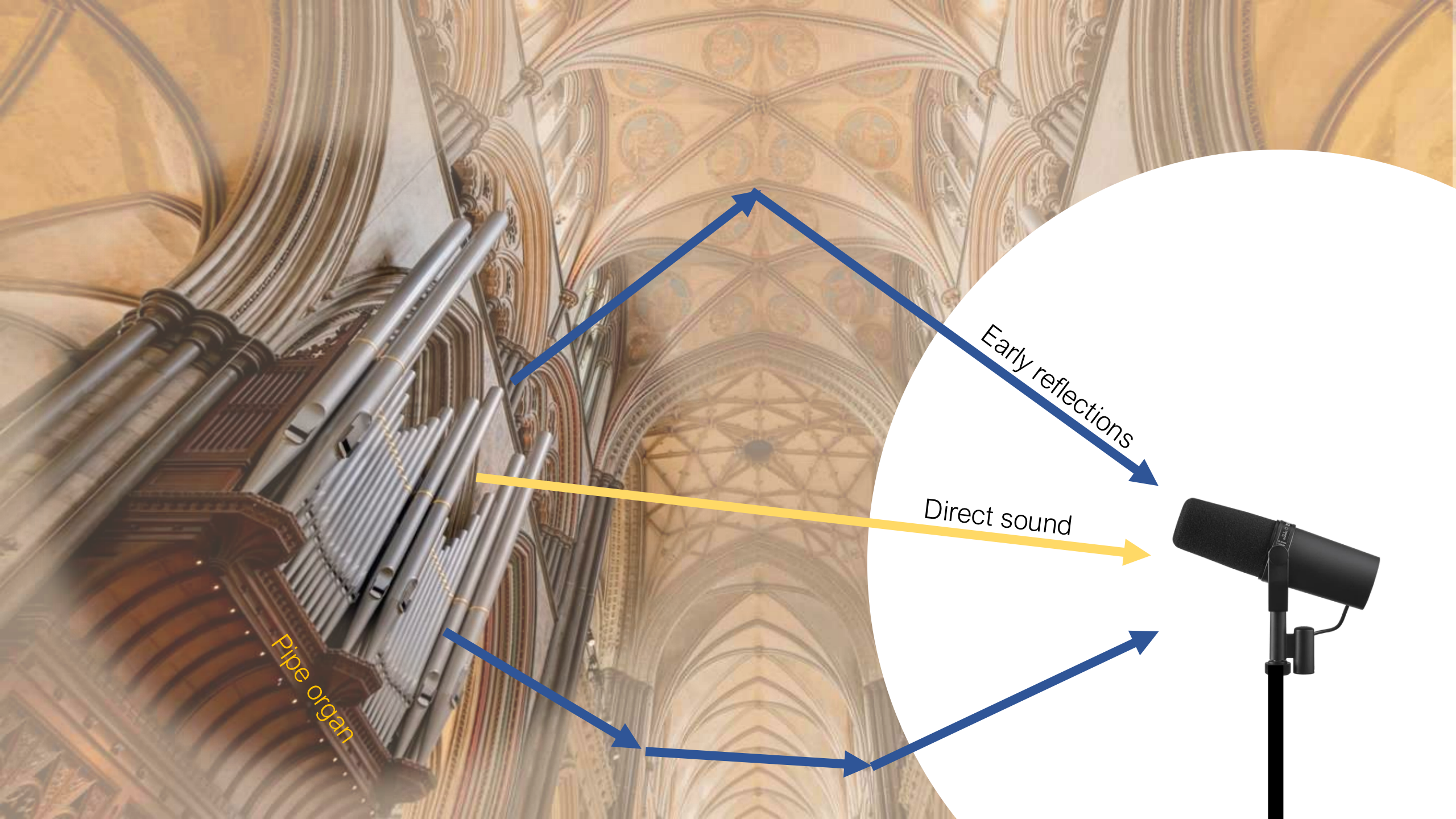
Reflected light

Direct light



Pipe organ

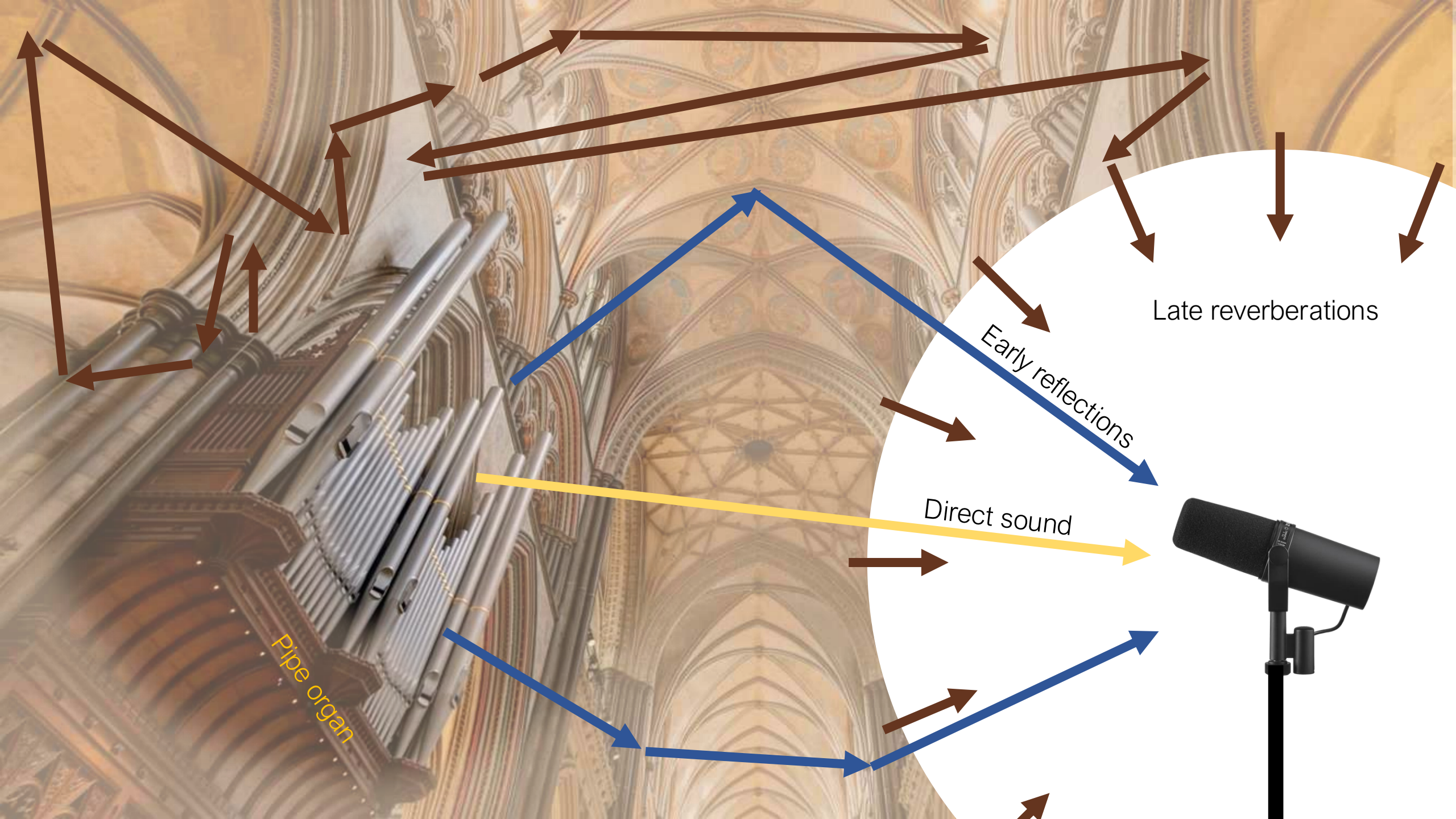
Direct sound



Pipe organ

Early reflections

Direct sound



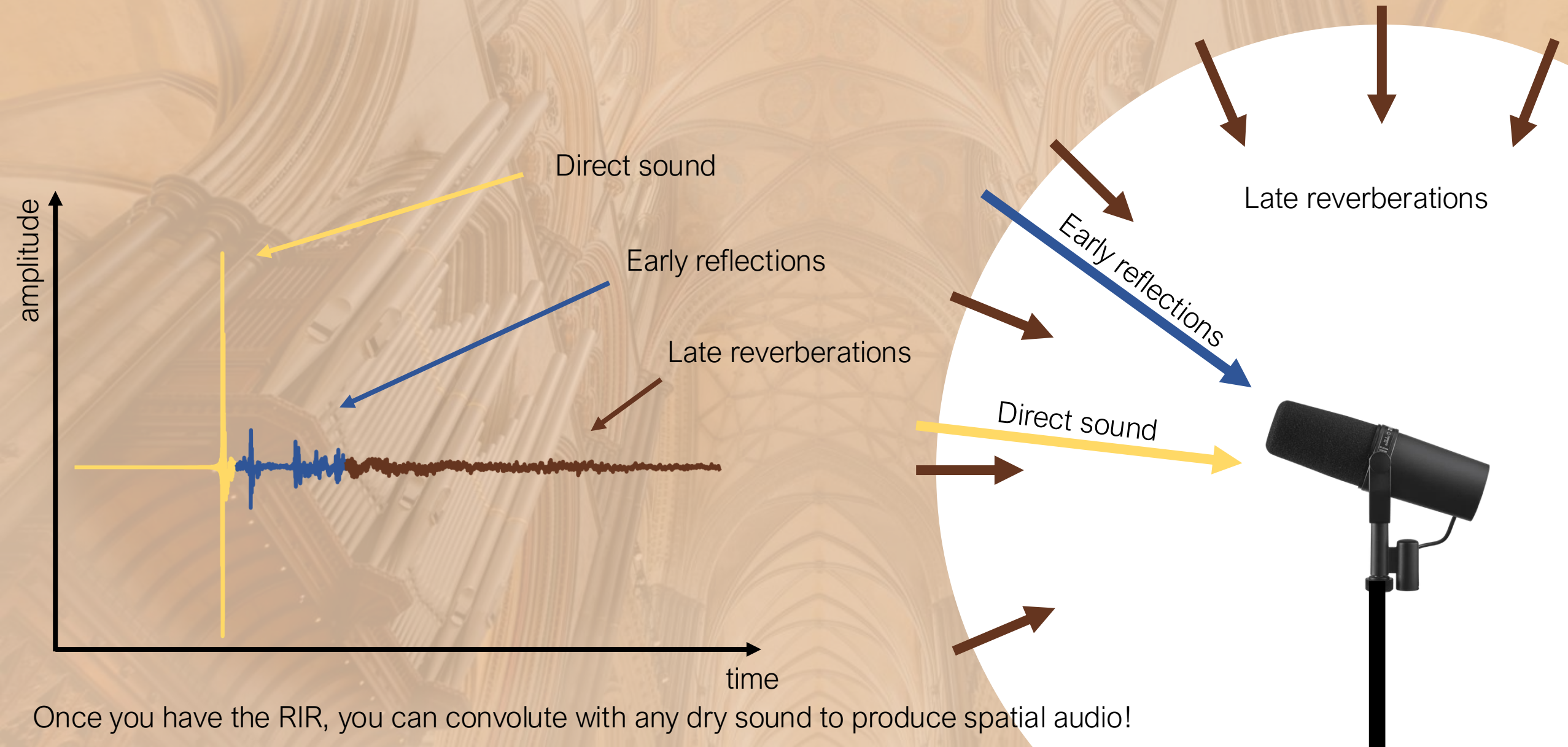
Pipe organ

Direct sound

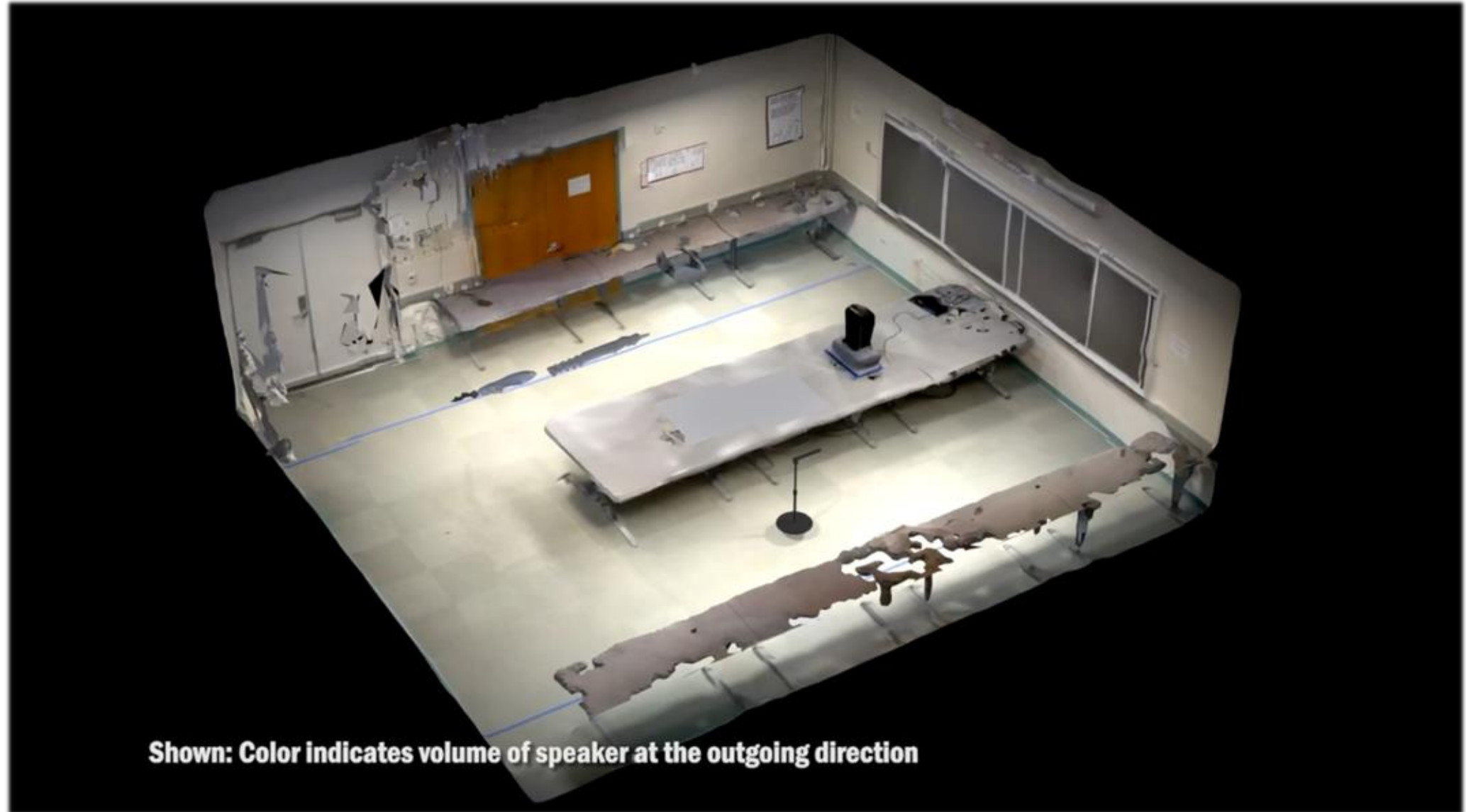
Early reflections

Late reverberations

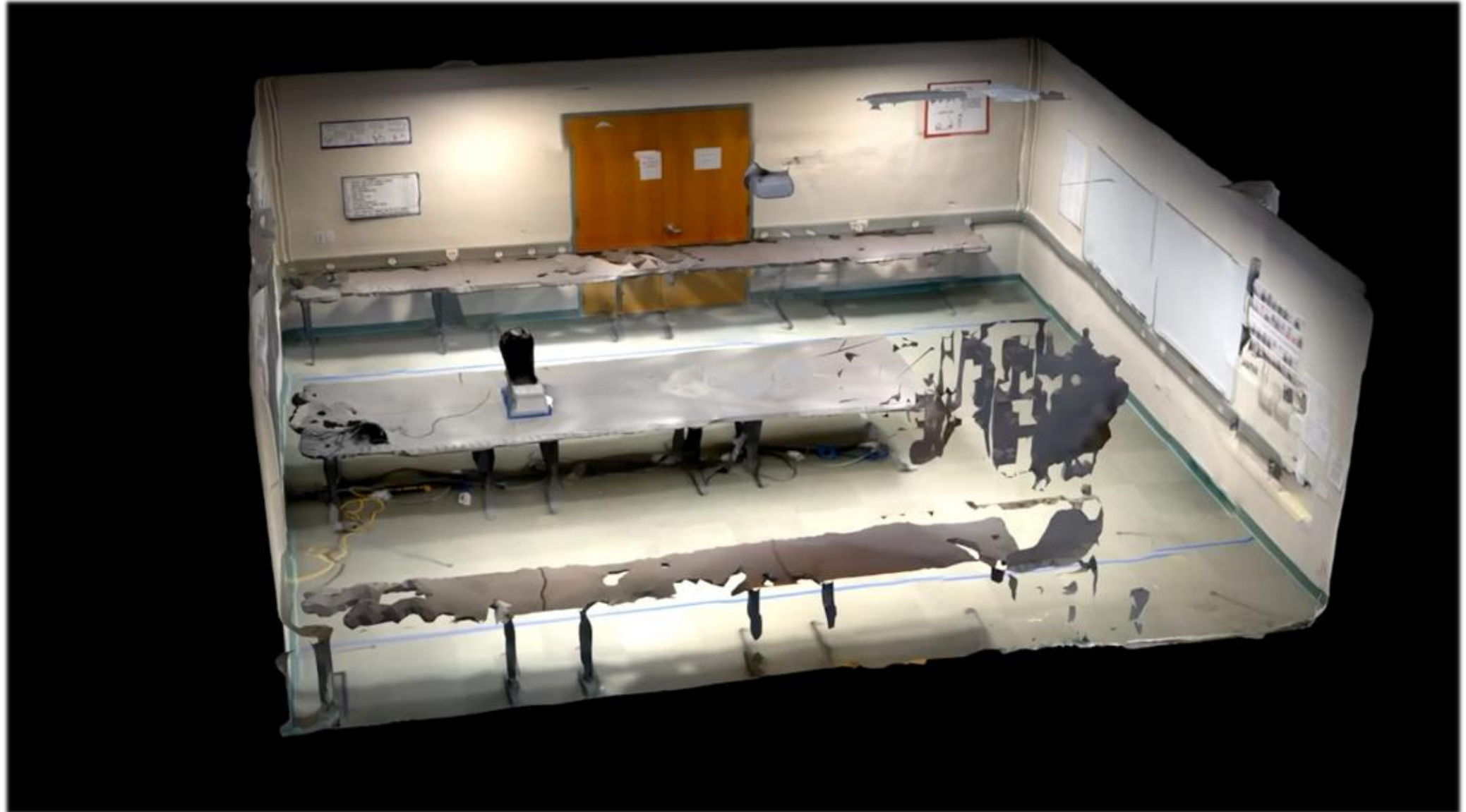
Room Impulse Response (RIR)



Differentiable Room Impulse Response Modeling



Differentiable Room Impulse Response Modeling



Prior Work on RIR Prediction

Traditional RIR Interpolation

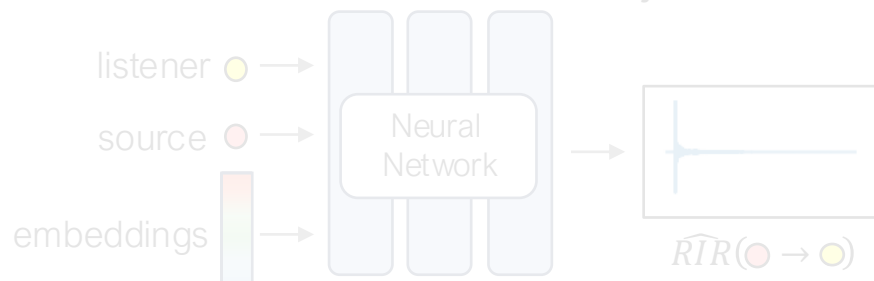
- Nearest Neighbor (NN)
- Linear Interpolation



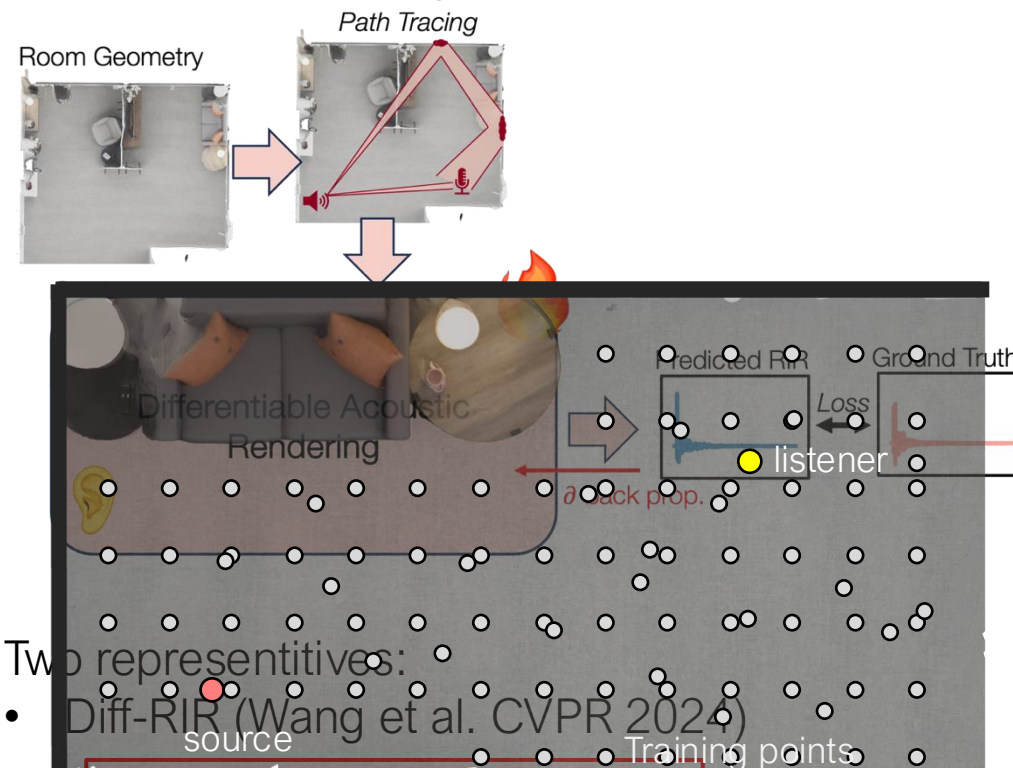
Need densely measured
ground-truth RIRs

Learning Based Methods

[Luo et al. 2022; Bhosale et al. 2024; Liang et al. 2023; Su et al. 2022; Ratnarajah et al. 2024]



Differentiable Physics-based Rendering



Two representatives:

- Diff-RIR (Wang et al. CVPR 2024)

- AVR (Lan et al. NeurIPS 2024)

Computationally expensive

High degree-of-freedom,
ill-posed inverse problem



Derong Jin

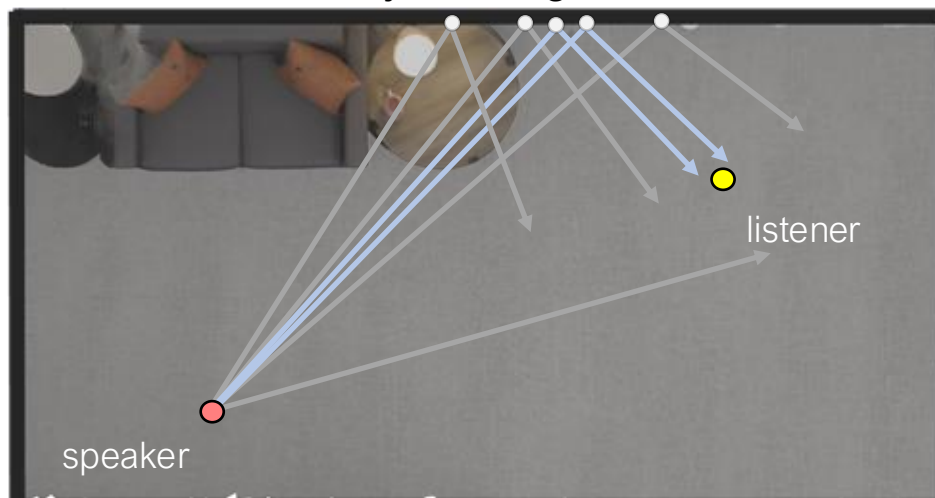
Our Key Observations

Differentiable Beam Tracing for efficient rendering

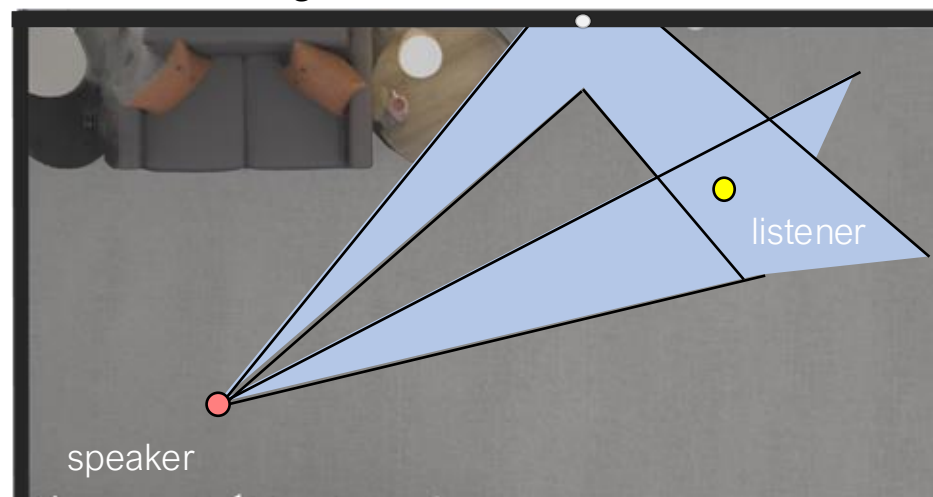
Multi-view vision priors for acoustic property learning

Differentiable Beam Tracing

Unidirectional Ray Tracing

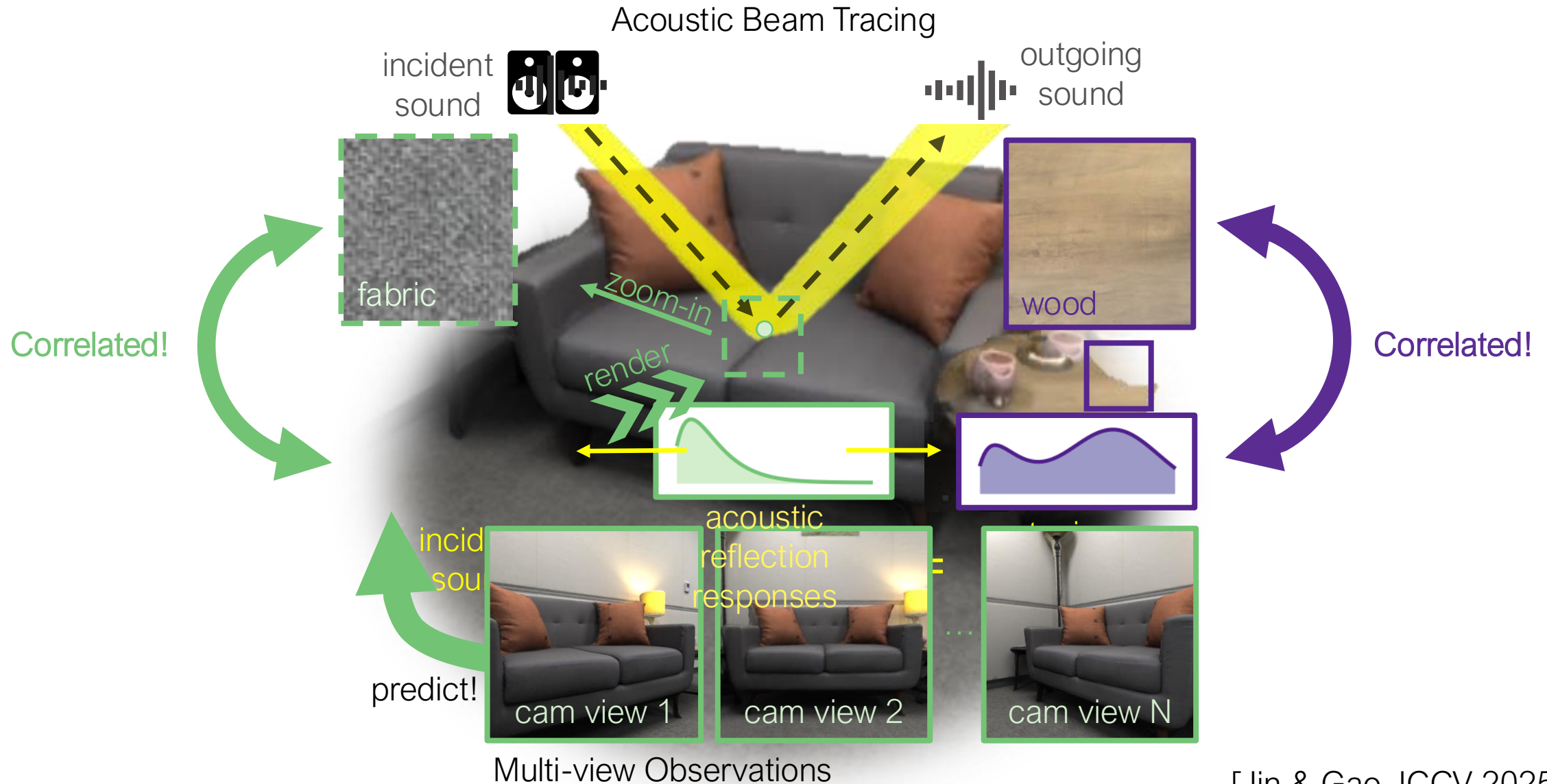


Beam Tracing

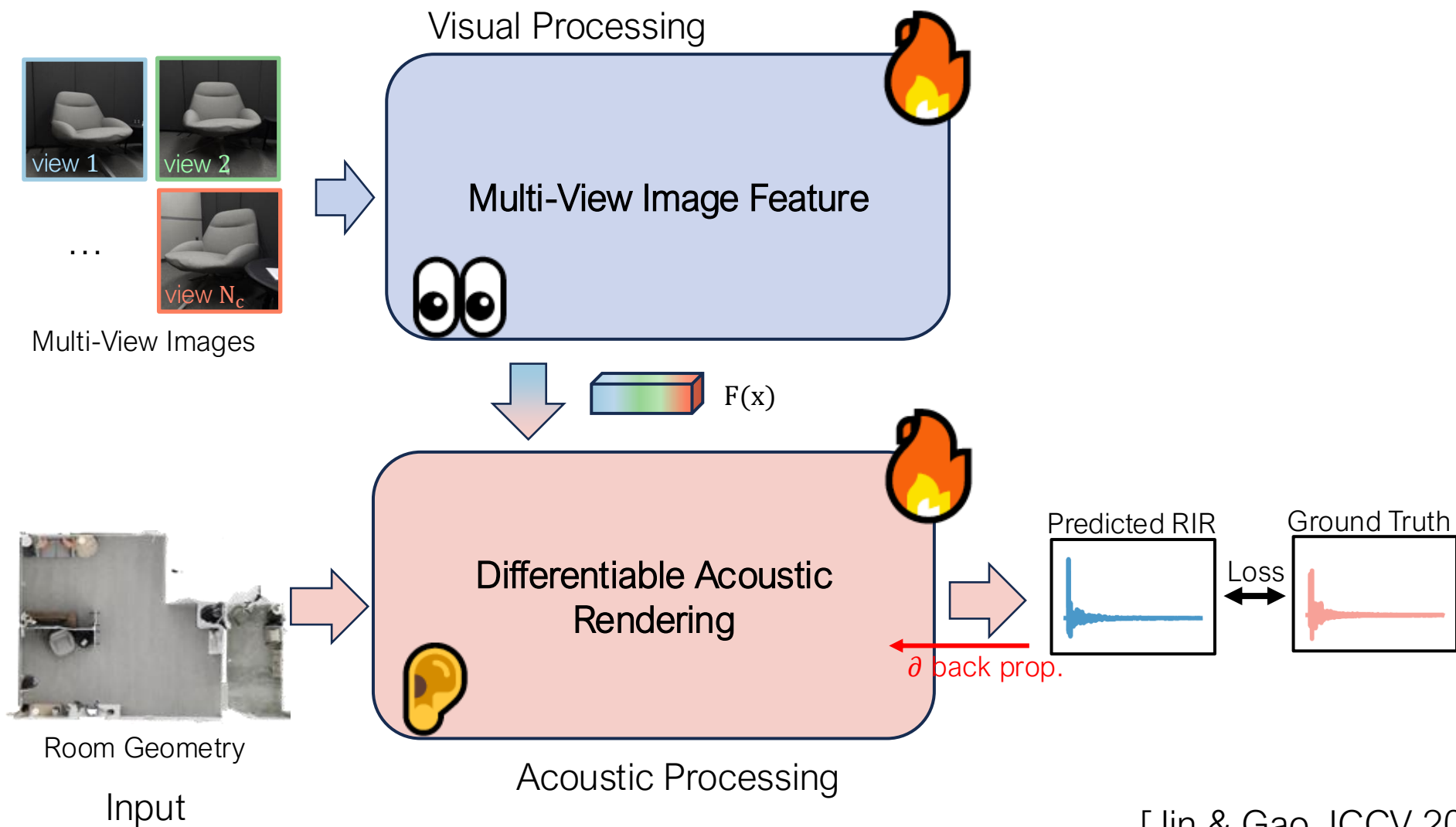


- many rays needed to find paths
- the same event is registered twice with different path length

Visual Cues Inform Differentiable Acoustic Learning



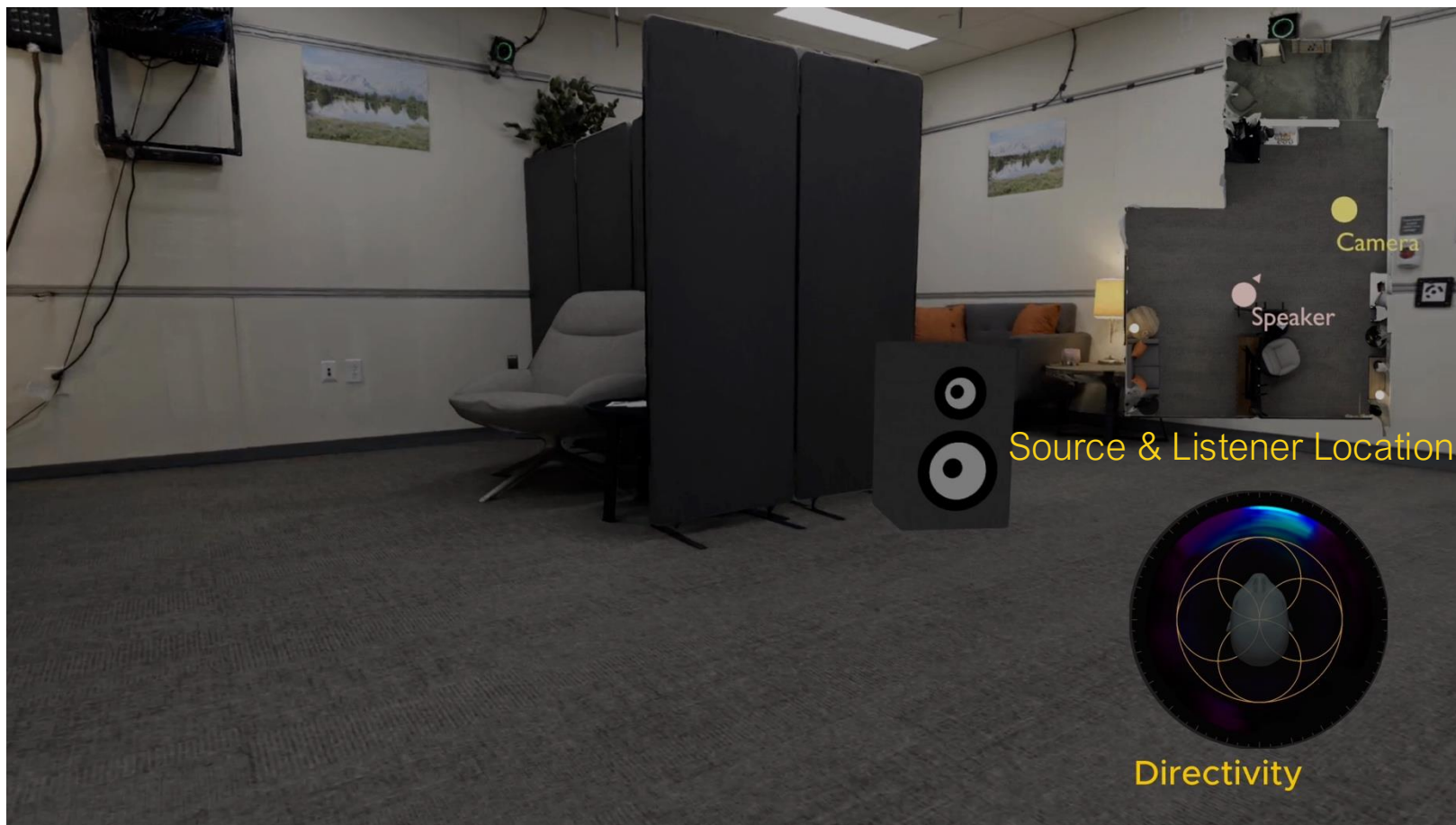
Audio-Visual Differentiable Acoustic Rendering



Demo Results



Dry
Sound





Camera



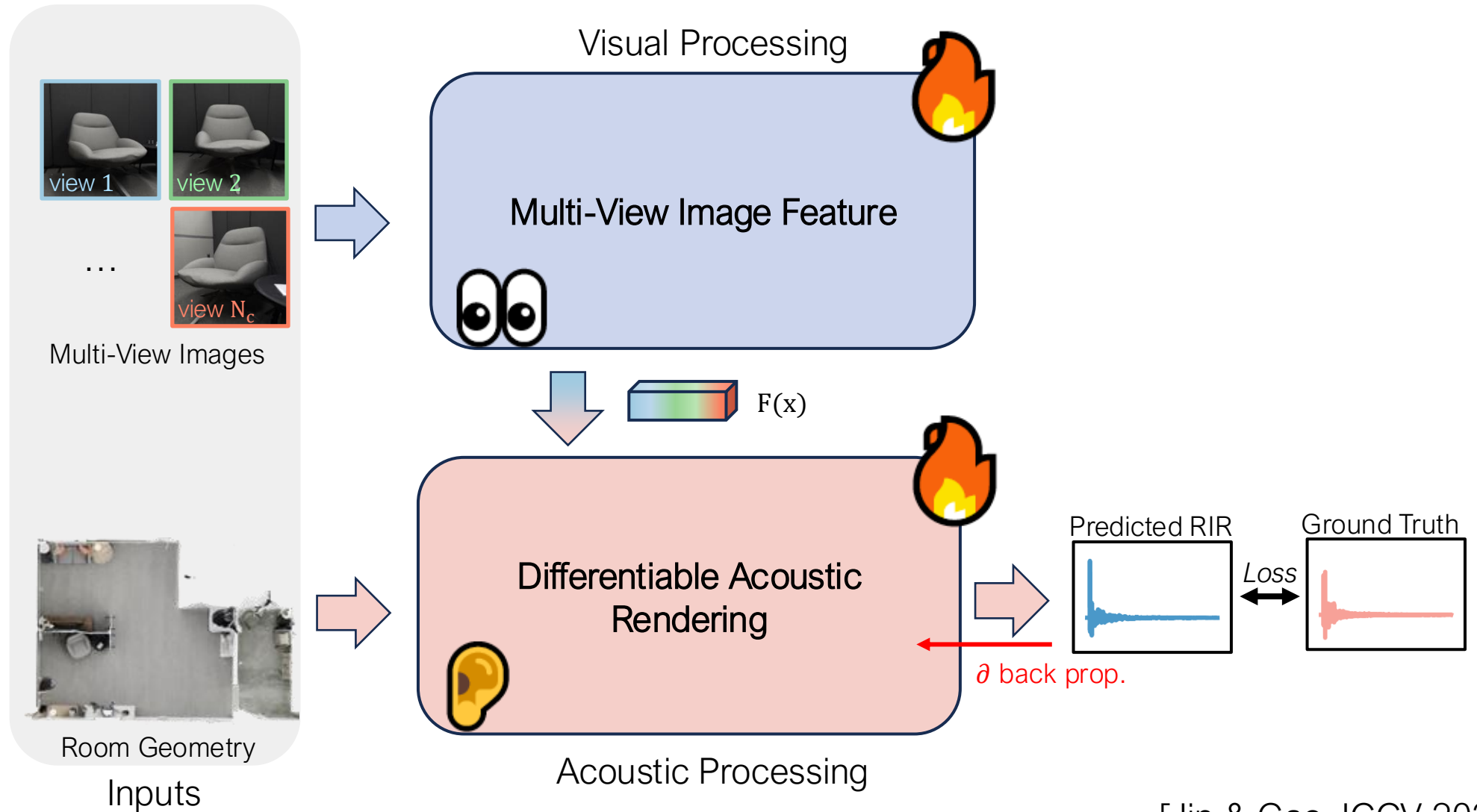
Speaker



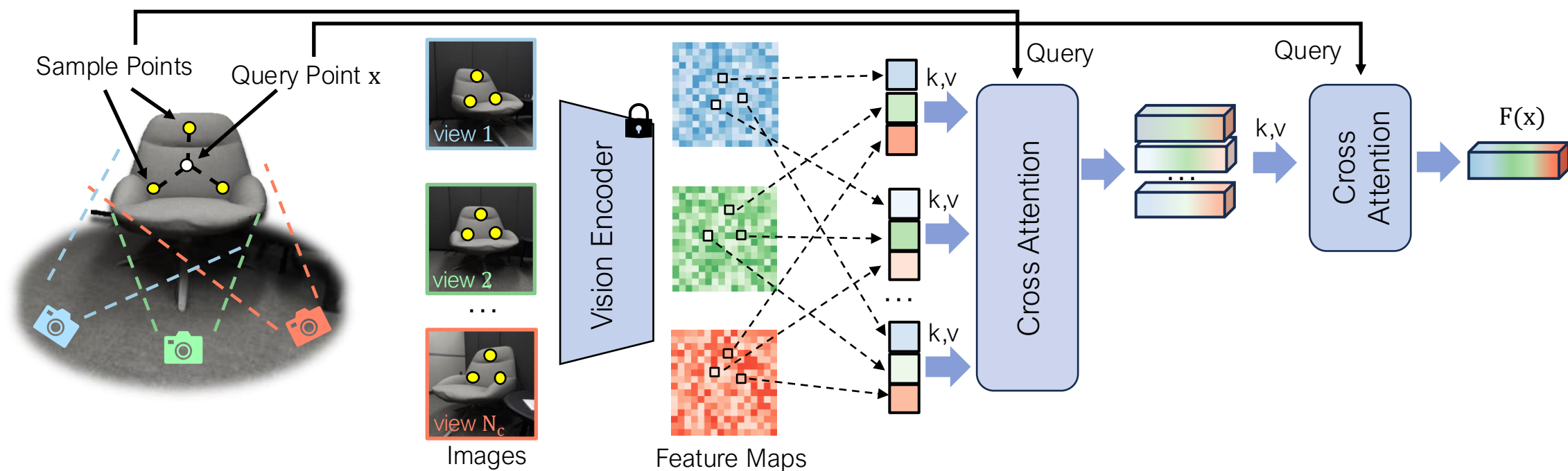
Directivity

[Jin & Gao, ICCV 2025 (Oral)]

Audio-Visual Differentiable Acoustic Rendering



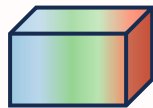
AV-DAR: Visual Processing



Multi-View Image Feature $F(x)$ evaluated at **any position** x on room surface.

AV-DAR: Audio Processing

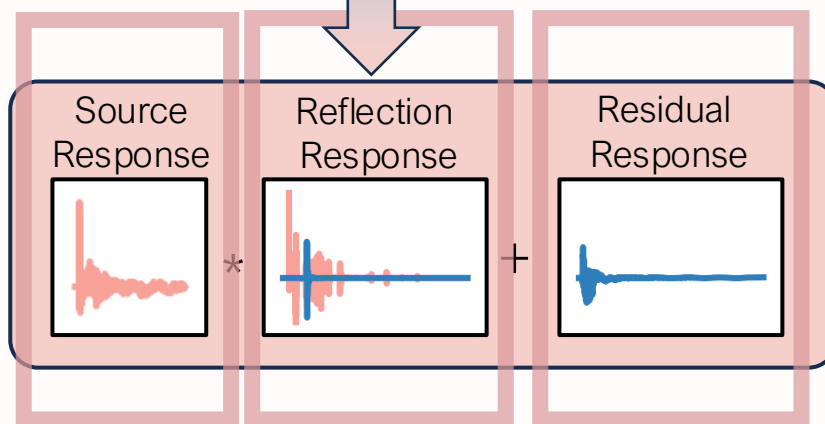
Multi-View Image Feature
from Visual Component



Source

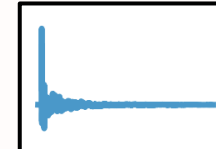
Listener

High-order reflections
and diffractions



Direct Sound and Specular Reflections

Predicted RIR

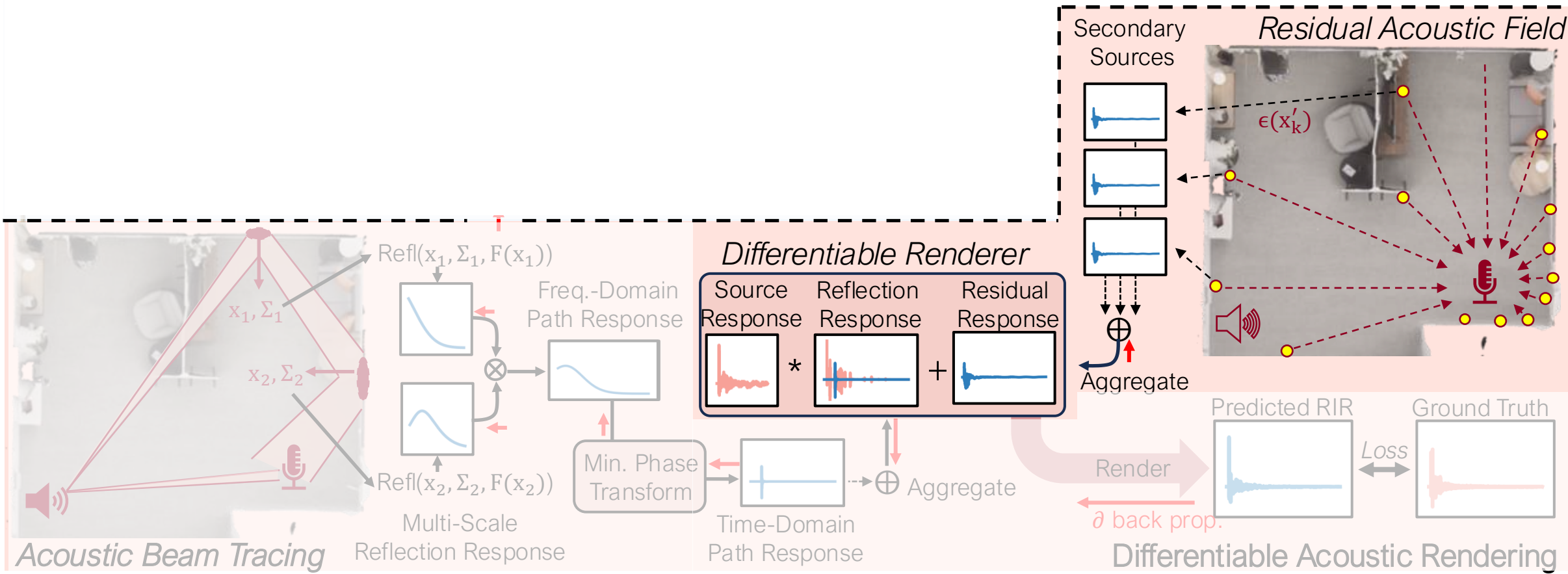


Differentiable Acoustic Rendering

Audio-Visual Differentiable Acoustic Rendering

Visual Processing

Acoustic Processing



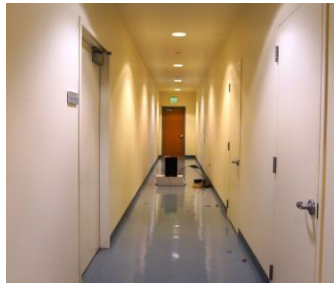
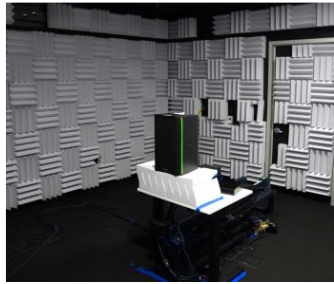
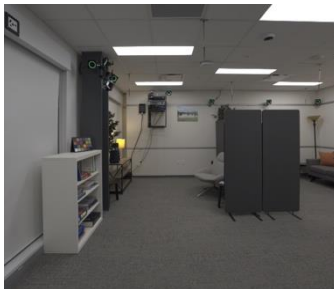
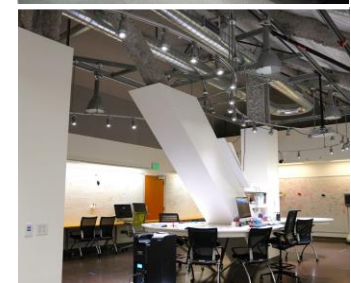
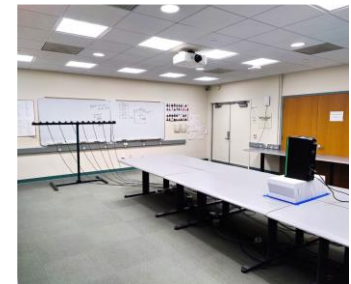
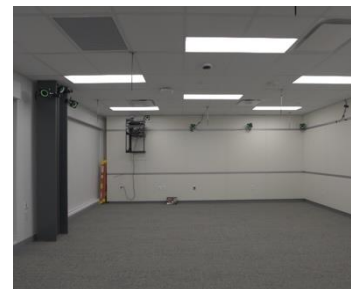
Demo 2

Source 1:
violins & flute

Quantitative Comparisons

RAF: Chen et al. CVPR'24

HAA: Wang et al. CVPR'24



		Scale	RAF-Empty				RAF-Furnished				T60 (%) ↓	
Method	Method		Loudness (dB) ↓	C50 (dB) ↓	EDT (ms) ↓	T60 (%) ↓	Loudness (dB) ↓	C50 (dB) ↓	EDT (ms) ↓	T60 (%) ↓		
Method	NAF++	1%	6.05	2.10	94.5	23.9	6.61	2.10	74.9	23.0	T60 (%) ↓	
	INRAS++	1%	3.69	2.59	100.3	23.5	2.96	2.61	92.6	25.0		
	AV-NeRF	1%	3.16	2.52	96.4	21.8	2.92	2.64	96.7	24.5		
	AVR	1%	3.00	2.19	87.3	24.1	2.97	2.33	72.3	17.9		
INRAS++	Ours	0.1%	3.14	1.81	86.6	16.9	2.45	1.98	80.1	15.2	7.4	
AV-NeRF	Ours	1%	2.50	1.42	56.2	10.7	1.68	1.29	47.4	9.61	9.5	
AVR											7.0	
Diff-RIR	2.24	2.42	39.7	1.75	2.23	18.5	1.87	1.56	44.9	1.32	3.13	6.8
Ours	0.99	1.02	24.3	0.98	1.44	10.8	1.11	1.45	31.9	0.85	1.15	6.3

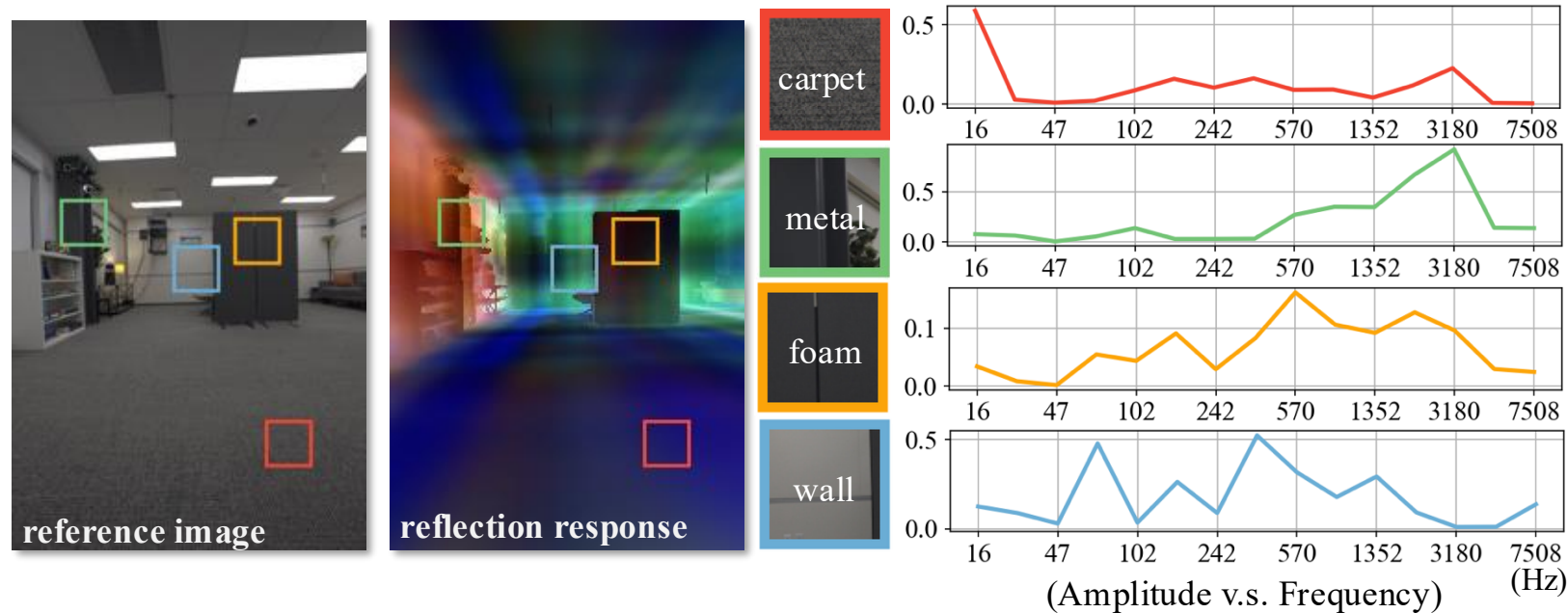
Best RIR prediction result across six real-world environments

[Wang et al. CVPR'24, Luo et al., NeurIPS'22, Su et al., NeurIPS'22,
Liang et al., NeurIPS'23, Lan et al., NeurIPS'24]

[Jin & Gao, ICCV 2025 (Oral)]

Visualizing the Learned Reflection Responses

- Encoding the
 - high-frequency response with **red**,
 - low-frequency response with **blue**:

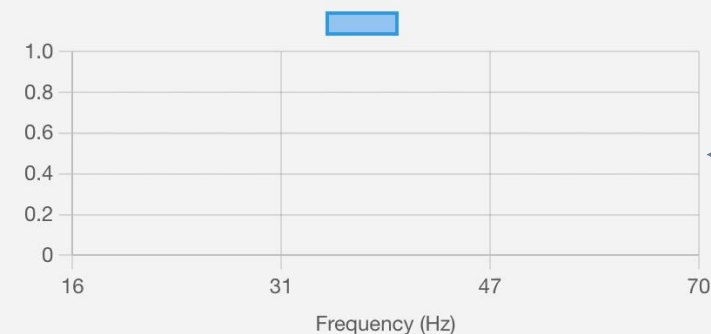


Visualizing the Learned Reflection Responses

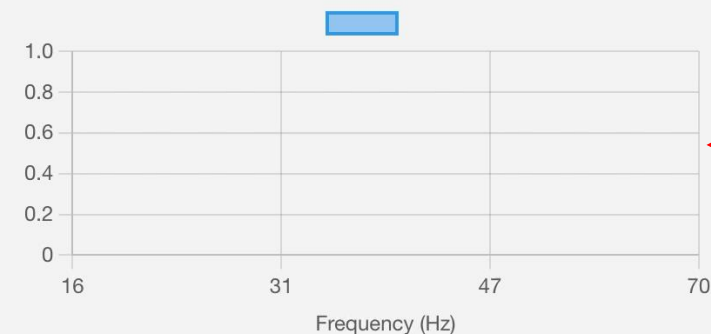
Reflection Response of Different Surfaces (Vision Helps!)



Reflection Response Values (Amplitude v.s. Frequency)



← w/ Vision



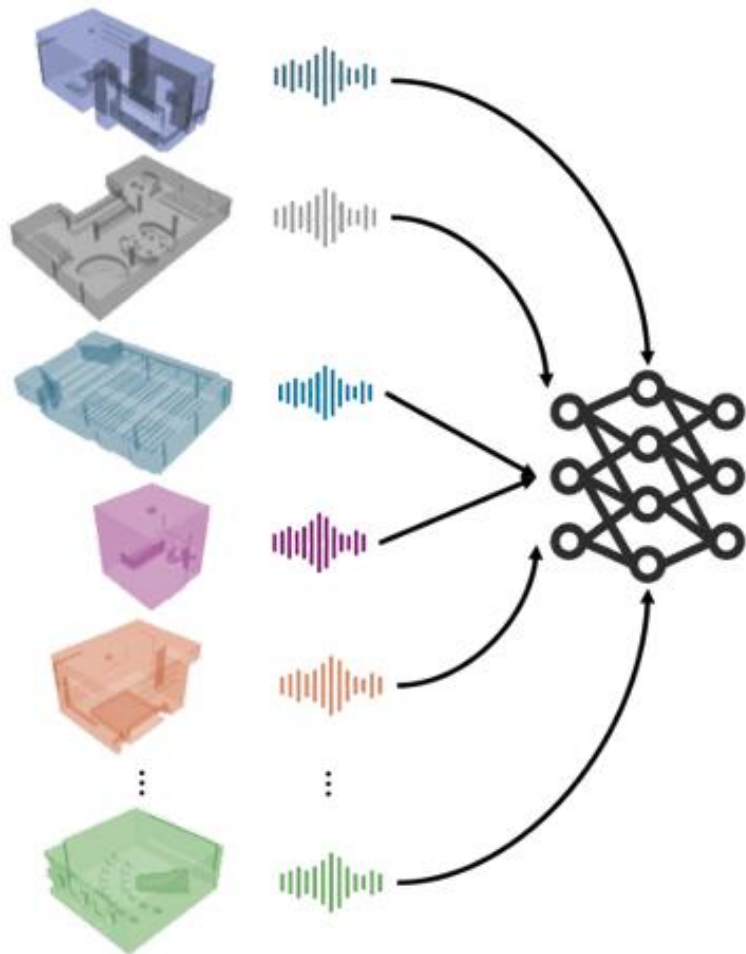
← w/o Vision



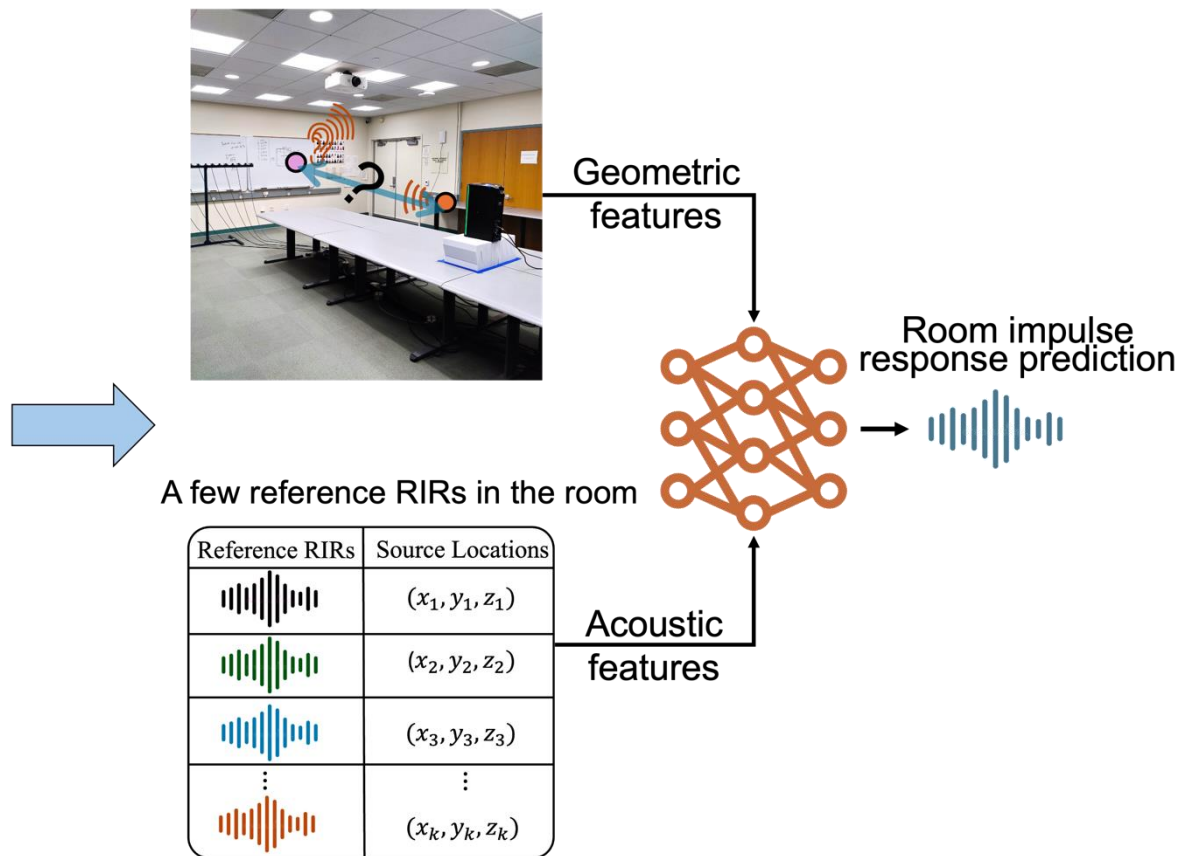
Xiulong Liu

Hearing Anywhere in Any Environment

Training environments



Inference anywhere in a new environment



A model trained on large number of rooms with diverse geometry and materials can accurately predict RIRs anywhere in a new environment.



Derong Jin



Xiyi Chen

From One Image to a 3D Audio-Visual Scene



- **Input:** One RGB image
- **Output:** 3D Audio-Visual Scene:
 - A *navigable* 3D visual scene
 - A spatial sound field that is semantically and geometrically aligned with the visual content

Demo Results



Input Image

Directivity



Map

Summary

Ruohan Gao
rhgao@umd.edu

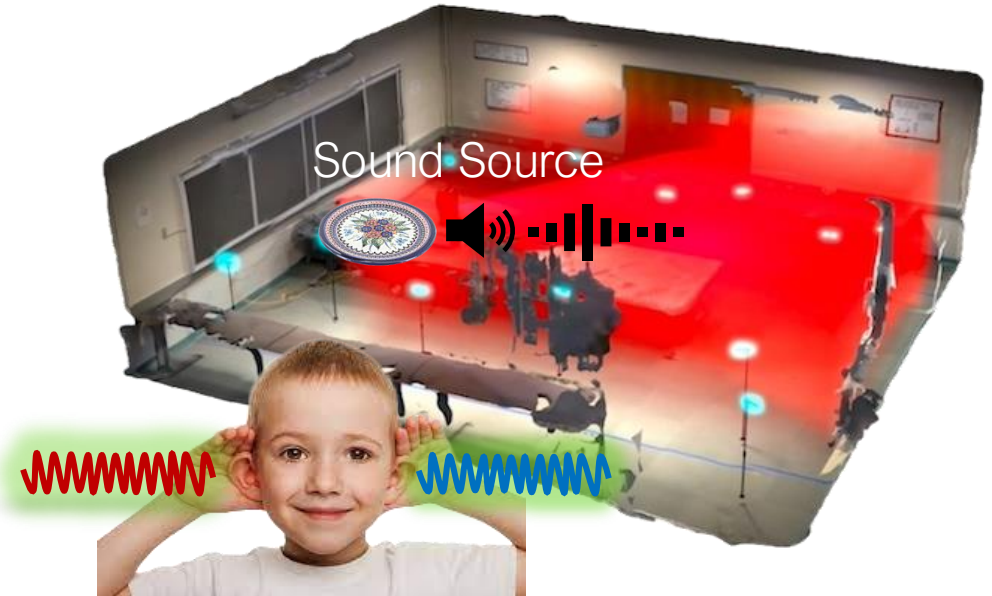
Audio-Visual Objects

Audio-Visual Space

Poster session:

**Thursday, Sept 10
4:30 PM – 6:30 PM**

Poster #16



1. Objects as Audio-Visual Modal Sound Field. Shao*, Wei*, et al. ECCV 2026.
2. Differentiable Room Acoustic Rendering with Multi-View Vision Priors. Jin and Gao. ICCV 2025 (Oral).
3. SonoWorld: From One Image to a 3D Audio-Visual Scene. Jin*, Chen*, et al. CVPR 2026.
4. ObjectFolder 2.0: A Multisensory Object Dataset for Sim2Real Transfer. CVPR 2022.
5. Hearing Anything Anywhere. Wang*, Sawata*, et al. CVPR 2024.
6. Hearing Anything in Any Environment. Liu et al. CVPR 2025.

<https://ruohangao.github.io/>